Econometrics Notes

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Last update: December 28, 2021

These notes were written by me when I was taking the Econometrics graduate course at PIMES - UFPE taught by Prof. Bladmir Carillo. It may contain errors and typos, so use it carefully and please contact me if you find any mistakes. In addition to the professor's lecture notes, I used as main reference the following textbooks:

- Angrist, Joshua David, and Jörn-Steffen Pischke. 2009. Mostly Harmless Econometrics: An Empiricist's Companion (MHE). Princeton: Princeton University Press.
- Cunningham, Scott. 2021. Causal Inference: The Mixtape. New Haven: Yale University Press. Available at https://mixtape.scunning.com/
- Greene, William H. 2008. Econometric Analysis. 6. ed. Upper Saddle River, NJ: Pearson, Prentice Hall.
- Hansen, Bruce. 2021. Econometrics. Available at https://www.ssc.wisc.edu/ bhansen/econometrics/

1 Regression Basics

1.1 Classical Linear Regression Model

We study the relationship between a *dependent variable* y and a set of *independent variables* x_1, ..., x_k:

$$y_i = x_{i1}\beta_1 + ... + x_{ik}\beta_k + \epsilon_i$$

where $i = 1, ..., n$ indexes observations and ϵ is an *error term* which aggregates factors not captured by the model. In matrix notation, we have a $n \times 1$ vector of dependent variables $\mathbf{Y}$, a $n \times K$ data matrix X, a $K \times 1$ vector of coefficients $\boldsymbol{\beta}$, and a $n \times 1$ vector of errors $\boldsymbol{\epsilon}$:

$$\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon} \tag{1.1}$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & ... & x_{1k} \\ x_{21} & y_{22} & ... & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & ... & x_{nk} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

1.1.1 Assumptions

- **A1. Linearity:** $y_i = x_{i1} + ... + x_{ik} + \epsilon_i$, $(i = 1, ..., n)$;
- **A2. No multicollinearity:** the rank of the matrix X is K (full rank), i.e., there is no perfect correlation between any of the independent variables (identification);
- **A3. Strict Exogeneity:** $\mathrm{E}[\epsilon_i|X] = 0$, $(i = 1, ..., n)$;
  - Strict exogeneity implies that *i.)* $\mathrm{E}[\epsilon_i] = 0$ and *ii.)* $\mathrm{E}[x_{jk}\epsilon_i] = 0$, $(i, j = 1, ..., n)$ and $(k = 1, ..., K)$.
- **A4. Spherical error variance:**
  - *Homoscedasticity:* $\mathrm{E}[\epsilon_i^2] = \sigma^2$, $(i = 1, ..., n)$;
  - *No autocorrelation:* $\mathrm{E}[\epsilon_i\epsilon_j] = 0$, $(i, j = 1, ..., n)$, $i \neq j$;
- **A5. Random Sample:** (y_i, x_i) is i.i.d across observations.

1.2 Conditional Expectation Function (CEF)

The *conditional expectation function* (CEF) for a dependent variable Y_i, given a $K \times 1$ vector of covariates X_i is the expectation (population average) of Y_i, with X_i held fixed: $\mathrm{E}[Y_i|X_i]$. Due to the LIE, we have:

$$\mathrm{E}[Y_i] = \mathrm{E}[\mathrm{E}[Y_i|X_i]]$$

Now, let ϵ_i be mean independent of X_i ($\mathrm{E}[\epsilon_i|X_i] = 0$), and hence uncorrelated to any function of X_i. We derive the *CEF decomposition property*:

$$Y_i = \mathrm{E}[Y_i|X_i] + \epsilon_i = \mathrm{CEF} + \epsilon_i$$

Hence, any random variable Y_i can be decomposed into a piece that is explained by X_i (the CEF) and a piece left over that is orthogonal to X_i. Moreover, the CEF is the best predictor of Y_i given X_i, in the sense that it solves a minimum means squared error (MMSE) problem. Let $m(X_i)$ be any function of X_i. The CEF solves

$$\mathrm{E}[Y_i|X_i] = \arg\min_{m(X_i)} \mathrm{E}[(Y_i - m(X_i))^2]$$

The CEF and its properties provide some reasons to consider the population regression 1.1 coefficients as parameters of interest. Namely,

- If the CEF is linear, it is equal to the population regression function;

- The function $X_i'\beta$ is the best linear predictor of Y_i given X_i (MMSE);
- The function $X_i'\beta$ provides the MMSE linear approximation to the CEF.

Therefore, regression provides the best linear predictor for the dependent variable in the same way that the CEF is the best *unrestricted* predictor. Also, even if the CEF is nonlinear, regression gives the best linear approximation to it.[1]

1.3 Useful results

Here we list some theorems and results extensively applied to derive econometric estimators' properties.

- Law of Large Numbers (LLN): sample moments converge in probability to the corresponding population moments, that is, the probability that the sample mean is close to the population mean approaches 1 with a large enough sample;
- Central Limit Theorem (CLT): normalized sample moments are asymptotically normally distributed;
- Slutsky's Theorem: let a_N be a random variable which converges in distribution and let b_N be a random variable that converges in probability to b. Then, $a_N + b_N$ has the same asymptotic distribution as $a_N + b$, and the same holds for $a_N b_N$ and $a_N b$.
- Continuous Mapping Theorem (CMT): The probability limit of any continuous function of a sample moment is the function evaluated at the corresponding population moment. Formally, $h(b_N) \xrightarrow{p} h(b)$ where $b_N \xrightarrow{p} b$ and $h(.)$ is continuous at b.
- Delta Method: Consider a vector-valued random variable with normal asymptotic distribution. Continuously differentiable scalar functions of this random variable are also asymptotically normal: $h(b_N)$ is normal, with covariance matrix $\nabla h(b)'\Omega\nabla h(b)$, where $b_N \xrightarrow{p} b$ and b_N has asymptotic covariance matrix Ω.

[1] For a detailed discussion on the CEF and proofs of the results, check MHE 3.1.

2 Ordinary Least Squares (OLS)

2.1 OLS estimator

The unknown parameters β and ϵ in the population regression 1.1 are never truly observed. We *estimate* them using sample data on $\mathbf{Y}$ and X:

$$\hat{y}_i = \mathbf{x_i}'\hat{\beta}$$

Importantly, we distinguish between model's *errors* ϵ_i and estimated *residuals* e_i:

$$\epsilon_i = y_i - \mathbf{x_i}'\beta$$

$$e_i = y_i - \hat{y}_i = y_i - \mathbf{x_i}'\hat{\beta}$$

$$y_i = \mathbf{x_i}'\beta + \epsilon_i = \mathbf{x_i}'\hat{\beta} + e_i$$

The OLS estimator of the β coefficient minimizes the sum of squared residuals:

$$S = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \mathbf{x_i}'\hat{\beta})^2$$

This optimization problem can be written in matrix form as

$$\begin{aligned} \min_{\beta} S = \mathbf{e}'\mathbf{e} &= (Y - X\beta)'(Y - X\beta) \\ &= Y'Y - 2Y'X\beta + \beta'X'X\beta \end{aligned}$$

From the first order condition, we obtain the solution for the OLS estimator:

$$\frac{\partial S}{\partial \beta} : -2X'Y + 2X'X\beta = 0$$

$$\hat{\beta}_{OLS} = (X'X)^{-1}X'Y \tag{2.1}$$

The full rank assumption **A2.** ensures that the matrix $(X'X)$ is invertible and positive definite, therefore the second order condition of the optimization problem is met. Thus, $\hat{\beta}_{OLS}$ is the unique solution for the minimization of the sum of squared residuals.[2]

2.2 Projection

We now define two useful matrices in linear regression. The vector of OLS residuals is

$$\begin{aligned} e = y - \hat{\beta}_{OLS} &= y - X(X'X)^{-1}X'Y \\ &= (I - X(X'X)^{-1}X)Y = MY \end{aligned}$$

The $n \times n$ matrix M is symmetric ($M = M'$) and idempotent ($M = M^2$). M is the *residual maker* of the regression of Y on X: $e = MY$. It follows that $MX = 0$. Analogously:

$$\hat{Y} = Y - e = (I - M)Y = X(X'X)^{-1}X'Y = PY$$

P is also symmetric and idempotent. It gives the *projection* of Y into the column space of X, i.e., the fitted values of the regression of Y on X: $\hat{y} = PY$. It follows that M and P are orthogonal ($MP = PM = 0$) and that $PX = X$. Finally,

$$\begin{aligned} Y = X\hat{\beta}_{OLS} + e &= \hat{Y} + e \\ Y = PY + MY &= \text{projection} + \text{residual} \end{aligned}$$

[2] Check Greene, page 22 for a detailed proof.

2.3 Partitioned Regression

Consider the bivariate model: $Y = X_1\beta_1 + X_2\beta_2 + \epsilon$. The *normal equations* are:

$$(1) : X_1'X_1\beta_1 + X_1'X_2\beta_2 = X_1'Y$$
$$(2) : X_2'X_1\beta_1 + X_2'X_2\beta_2 = X_2'Y$$

Rearranging (1) yields

$$\hat{\beta}_1 = (X_1'X_1)^{-1}X_1'(Y - X_2\beta_2) \quad (*)$$

Note that, if X_1 and X_2 are orthogonal ($X_1'X_2 = 0$), $\hat{\beta}_1$ is simply $(X_1'X_1)^{-1}X_1'Y$, i.e., the coefficient of a univariate regression of Y on X_1. Now, plugging (*) in (2):

$$X_2'X_1(X_1'X_1)^{-1}X_1'Y - X_2'X_1(X_1'X_1)^{-1}X_1'X_2\beta_2 + X_2'X_2\beta_2 = X_2'Y$$
$$\implies \hat{\beta}_2 = [X_2'(I - X_1(X_1'X_1)^{-1}X_1')X_2]^{-1}[X_2'(I - X_2(X_1'X_1)^{-1}X_1')Y]$$
$$= (X_2'M_1X_2)^{-1}(X_2'M_1Y)$$

where M_1 is the residual maker of the regression of Y on X_1. Therefore, we have:

$$\hat{\beta}_1 = (X_1'M_2X_1)^{-1}(X_1'M_2Y)$$
$$\hat{\beta}_2 = (X_2'M_1X_2)^{-1}(X_2'M_1Y)$$

2.3.1 Frisch-Waugh-Lovell Theorem (FWL)

Since M_2 is idempotent, we can rewrite the previous result for $\hat{\beta}_1$ as

$$\hat{\beta}_1 = (X_1'M_2M_2X_1)^{-1}(X_1'M_2M_2Y)$$
$$= (\tilde{X_1}'\tilde{X_1})^{-1}(\tilde{X_1}'\tilde{e_2})$$

where $\tilde{X_1} = M_2X_1$ and $\tilde{e_2} = M_2Y$. Thus, the FWL theorem provides us a way to obtain the OLS coefficient of X_1 by *netting out* the effect of X_2. The theorem indicates three steps:

1. Regress Y on X_2 and save residuals $\tilde{e_2}$;
2. Regress X_1 on X_2 and save residuals $\tilde{X_1}$;
3. Regress $\tilde{e_2}$ on $\tilde{X_1}$ and obtain OLS estimates for $\hat{\beta}_1$ and e.

More generally, what happens when we add a variable to the regression? Let e denote the residuals of a regression of Y on X and u denote the residuals when Z is included as well. We have

$$e = Y - X\hat{\beta}$$
$$u = Y - Xd - Zc$$

The coefficient c can be obtained via FWL:

$$c = (Z'MZ)^{-1}(Z'MY) = (\tilde{Z}'\tilde{Z})^{-1}\tilde{Z}Y \quad (*)$$

We obtain d from the partitioned regression:

$$d = (X'X)^{-1}X'(Y - Zc) = \hat{\beta} - (X'X)^{-1}X'Zc \quad (**)$$

Now, inserting (**) in the expression for u:

$$u = Y - X\hat{\beta} + (X'X)^{-1}X'Zc - Zc$$
$$= e - MZc = e - \tilde{Z}c$$
$$\implies u'u = e'e + c^2(\tilde{Z}'\tilde{Z}) - 2c\tilde{Z}'e$$

But $e = MY = \tilde{Y}$ and $\tilde{Z}'e = \tilde{Z}'\tilde{Y} = c(\tilde{Z}'\tilde{Z})$, in which the last equality comes from (*). Thus,

$$u'u = e'e + c^2 - 2c^2(\tilde{Z}'\tilde{Z})$$
$$= e'e - c^2(\tilde{Z}'\tilde{Z}) \leq e'e$$

Therefore, adding a variable to the regression leads to a sum of squared residuals *smaller or equal to* the original regression's sum of squared residuals.

2.4 Goodness of fit

We now describe a measure of fit that embodies the fitting criterion (sum of squared residuals) and the covariation of y and x. The total variation in y, or *total sum of squares* (SST), is the sum of squared deviations from its mean:

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

If the regression contains a constant term, then the residuals will sum to zero and the mean of the predicted values of y_i will equal the mean of the actual values (see Section 2.5.1). Using this, for each observation we have

$$y_i - \bar{y} = \hat{y}_i + e_i - \bar{y} = (x_i - \bar{x})'\hat{\beta} + e_i$$

For the full set of observations, we have

$$M^0 Y = M^0 X\hat{\beta} + M^0 e$$

where M^0 is the matrix that transforms observations into deviations from sample means. Since residuals have mean zero, $M^0 e = e$ and $e' M^0 X = e' X = 0$, thus

$$Y' M^0 Y = \hat{\beta}' X' M^0 X \hat{\beta} + e' e$$

Therefore, the total sum of squares SST equals the *explained sum of squares*, $SSE = \sum_n (\hat{y}_i - \bar{y})^2$, plus the *sum of squared residuals*, $SSR = \sum_n e_i^2$:

$$SST = SSE + SSR \tag{2.2}$$

With this we define our measure of fit, R^2:

$$R^2 = \frac{SSE}{SST} = \frac{\hat{\beta}' X' M^0 X \hat{\beta}}{Y' M^0 Y} = 1 - \frac{SSR}{SST} = 1 - \frac{e'e}{Y' M^0 Y} \tag{2.3}$$

The coefficient of determination R^2 ranges from 0 to 1 and measures the proportion of the variation in y that is accounted for by variation in the regressors X.

2.4.1 Adjusted R^2

One problem with R^2 is that it will never decrease when another variable is added to the regression, which can be misleading. The *adjusted* R^2 (for degreees of freedom) incorporates a penalty for those results, and may be even negative:

$$\bar{R}^2 = 1 - \frac{e'e/(n-K)}{Y' M^0 Y/(n-1)} \tag{2.4}$$

The connection between R^2 and $\bar{R}^2$ is given by

$$\bar{R}^2 = 1 - \frac{n-1}{n-K}(1 - R^2)$$

2.5 OLS Properties

2.5.1 Algebraic aspects

The normal equations are

$$X' X\hat{\beta} - X' Y = -X'(Y - X\hat{\beta}) = -X' e = 0$$

Hence, for every column x_k, $x_k' e = 0$. If the first column of X is a columns of $1's$ – i.e., we add a constant term to the regression – there are three implications:

1. Since $x_1' e = \mathbf{i}' e = \sum_i e_i = 0$, the OLS residuals sum to zero;
2. The first normal equation implies that $\bar{y} = \bar{x}'\hat{\beta}$, thus the regression line passes through the point of means of the data;
3. From point 1, $\hat{y} = X\hat{\beta}$, hence the mean of the fitted values equals the mean of the actual values.

2.5.2 Finite Sample Properties

The OLS estimator is *unbiased*:

$$\begin{aligned}\hat{\beta}_{OLS} &= (X'X)^{-1}X'Y = (X'X)^{-1}X'(X\beta + \epsilon) \\ &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\epsilon \\ &= \beta + (X'X)^{-1}X'\epsilon \\ \implies \mathrm{E}[\hat{\beta}_{OLS}|X] &= \beta + \mathrm{E}[(X'X)^{-1}X'\epsilon|X]\end{aligned}$$

From **A3.**, $\mathrm{E}[\epsilon|X] = 0$. Thus:

$$\mathrm{E}[\hat{\beta}_{OLS}|X] = \beta$$

Finally, using the Law of Iterated Expectations:

$$\mathrm{E}[\hat{\beta}_{OLS}] = \mathrm{E}[\mathrm{E}[\hat{\beta}_{OLS}|X]] = \mathrm{E}_x[\beta]$$

$$\mathrm{E}[\hat{\beta}_{OLS}] = \beta \tag{2.5}$$

From 2.5 and $\hat{\beta}_{OLS} = \beta + (X'X)^{-1}X'\epsilon$ we can obtain the variance of the OLS estimator:

$$\begin{aligned}Var(\hat{\beta}_{OLS}|X) &= \mathrm{E}[(\hat{\beta}_{OLS} - \beta)(\hat{\beta}_{OLS} - \beta)'] \\ &= \mathrm{E}[(X'X)^{-1}X'\epsilon\epsilon'X(X'X)^{-1}|X] \\ &= (X'X)^{-1}X'\mathrm{E}[\epsilon\epsilon'|X]X(X'X)^{-1}\end{aligned}$$

From the homoscedasticity assumption (**A4.**), we have $\mathrm{E}[\epsilon\epsilon'|X] = \sigma^2$. Therefore:

$$Var(\hat{\beta}_{OLS}|X) = \sigma^2(X'X)^{-1} \tag{2.6}$$

We now derive the *Gauss-Markov Theorem*: in the classical linear regression model, the OLS estimator is the minimum variance linear unbiased estimator (MVLUE) of β.

Let $\tilde{\beta} = CY$ be a linear unbiased estimator of β other than $\hat{\beta}_{OLS}$. We shall prove that $\tilde{\beta}$'s variance is not smaller than the variance of the OLS estimator. If $\tilde{\beta}$ is unbiased, then

$$\mathrm{E}[CY|X] = \mathrm{E}[CX\beta + C\epsilon|X] = \beta \quad (*)$$

From the classical model's assumptions, (*) implies that CX must be equal to I. From the expression for $\hat{\beta}_{OLS}$'s variance 2.6, we can obtain $Var(\tilde{\beta}|X)$ by replacing $(X'X)^{-1}X'$ for C:

$$Var(\tilde{\beta}|X) = \sigma^2 CC' \quad (**)$$

Now, let $D = C - (X'X)^{-1}X'$, so that $DY = \tilde{\beta} - \hat{\beta}_{OLS}$. The expression (**) becomes

$$Var(\tilde{\beta}|X) = \sigma^2[(D + (X'X)^{-1}X')(D + (X'X)^{-1}X')']$$

But, $CX = DX + (X'X)^{-1}X'X = I$. Hence, DX must be equal to zero. Therefore:

$$\begin{aligned}Var(\tilde{\beta}|X) &= \sigma^2(X'X)^{-1} + \sigma^2 DD' \\ &= Var(\hat{\beta}_{OLS}|X) + \sigma^2 DD' \geq Var(\hat{\beta}_{OLS}|X)\end{aligned}$$

Since a quadratic form in DD' is $q'DD'q = z'z \geq 0$, the conditional covariance matrix of any $\hat{\beta}$ equals that of $\hat{\beta}_{OLS}$ plus a non-negative definite matrix, establishing the result.

To test hypothesis about β or to form confidence intervals, we need to estimate the covariance matrix 2.6. Thus, we seek an estimator of the error's variance σ^2. Since $\sigma^2 = \epsilon'\epsilon$ and the residuals $\mathbf{e}$ are estimates of the errors ϵ, an estimator of the error's variance will be based on the sum of squared residuals:

$$\begin{aligned}e &= MY = M[X\beta + \epsilon] = M\epsilon \\ e'e &= \epsilon' M\epsilon \\ \mathrm{E}[e'e|X] &= \mathrm{E}[\epsilon' M\epsilon|X]\end{aligned}$$

The scalar $\epsilon' M \epsilon$ is a 1×1 matrix, so it is equal to its trace. By cyclic permutations:[3]

$$\mathrm{E}[tr(\epsilon' M \epsilon)|X] = \mathrm{E}[tr(M \epsilon \epsilon')|X]$$

Since M is a function of X:

$$tr(M\mathrm{E}[\epsilon'\epsilon|X]) = tr(M\sigma^2 I) = \sigma^2 tr(M)$$

Now, the trace of M is

$$tr[I_n - X(X'X)^{-1}X'] = tr(I_n) - tr(I_k) = n - k$$
$$\implies \mathrm{E}[e'e|X] = (n-K)\sigma^2$$

Therefore, an unbiased estimator of σ^2 is

$$s^2 = \frac{e'e}{(n-K)} \tag{2.7}$$

Summary of finite sample properties of OLS:

- **FS1.** $\mathrm{E}[\hat{\beta}_{OLS}] = \beta$;
- **FS2.** $Var(\hat{\beta}_{OLS}|X) = \sigma^2(X'X)^{-1}$;
- **FS3.** $\hat{\beta}_{OLS}$ is MVLUE (Gauss-Markov);
- **FS4.** $\mathrm{E}[s^2|X] = \mathrm{E}[s^2] = \sigma^2$;
- **FS5.** $Cov(\hat{\beta}_{OLS}, e|X) = \mathrm{E}[(\hat{\beta}_{OLS} - \beta)e|X] = \mathrm{E}[(X'X)^{-1}X'\epsilon\epsilon' M|X] = 0$;

2.5.3 Asymptotic properties of OLS

To establish consistency of $\hat{\beta}_{OLS}$ in large samples we make two further assumptions: *i.*) (x_1, ϵ_i) is a sequence of independent observations; *ii.*) $p\lim_{n\to\infty} \frac{1}{n}X'X = Q$, is a positive definite matrix. Thus, we now have

$$\hat{\beta}_{OLS} = (X'X)^{-1}X'(x\beta) + (X'X)^{-1}X'\epsilon$$
$$= \beta + \left(\frac{X'X}{n}\right)^{-1}\left(\frac{X'\epsilon}{n}\right)$$
$$\implies p\lim_{n\to\infty} \hat{\beta}_{OLS} = \beta + Q^{-1} p\lim_{n\to\infty}\left(\frac{X'\epsilon}{n}\right)$$

Now, let $\bar{w} = \left(\frac{X'\epsilon}{n}\right)$. From assumption **A3.**:

$$\mathrm{E}[w_i] = \mathrm{E_x}[\mathrm{E}[w_i|X_i]] = \mathrm{E_x}[X_i\mathrm{E}[\epsilon_i|X_i]] = 0$$
$$\implies \mathrm{E}[\bar{w}] = 0$$

We get the variance of $\bar{w}$ using the ANOVA Theorem:[4]

$$Var(\bar{w}) = \mathrm{E}[Var(\bar{w}|X)] + Var(\mathrm{E}[\bar{w}|X])$$
$$= \mathrm{E}[Var(\bar{w}|X)] = \mathrm{E}[\bar{w}\bar{w}'|X]$$
$$= \frac{1}{n}X'\mathrm{E}[\epsilon\epsilon'|X]X\frac{1}{n} = \left(\frac{\sigma^2}{n}\right)\left(\frac{X'X}{n}\right)$$
$$\implies \lim Var(\bar{w}) = 0 \times Q = 0$$

Since $\bar{w}$ has mean zero and its variance converges to zero:

$$p\lim_{n\to\infty} \bar{w} = p\lim_{n\to\infty}\left(\frac{X'\epsilon}{n}\right) = 0$$

[3] See Greene, page 971.
[4] See MHE, page 33.

Therefore, $\hat{\beta}_{OLS}$ is consistent:

$$p \lim_{n\to\infty} \hat{\beta}_{OLS} = \beta$$

We now derive the asymptotic distribution of $\hat{\beta}_{OLS}$. If $\{\epsilon_i\}$ are idependently distributed with mean zero (**A3.**) and finite variance σ^2 (**A4.**), and x_{ik} is such that the *Grenander conditions*[5] are met:

- *i.*) for each column in X, $\mathbf{x_k}$, $\lim_{n\to\infty} x_k' x_k = +\infty$;
- *ii.*) $\lim_{n\to\infty} \frac{x_{ik}^2}{x_k' x_k} = 0$, $\forall i$;
- *iii.*) let R_n be the correlation matrix of the columns in X, then $\lim_{n\to\infty} R_n = C$, a positive definite matrix.

Then, $\hat{\beta}_{OLS}$ asymptotically has a normal distribution:[6]

$$\hat{\beta}_{OLS} \sim \mathrm{N}\left[\beta, \frac{\sigma^2}{n} Q^{-1}\right]$$

We also establish consistency of s^2. Since $e'e = \epsilon' M \epsilon$,

$$s^2 = \frac{\epsilon' M \epsilon}{(n-K)} = \frac{[\epsilon'\epsilon - \epsilon' X (X'X)^{-1} X' \epsilon]}{(n-K)} = \frac{n}{(n-K)}\left[\frac{\epsilon'\epsilon}{n} - \left(\frac{\epsilon' X}{n}\right)\left(\frac{X'X}{n}\right)^{-1}\left(\frac{X'\epsilon}{n}\right)\right]$$

As $n \to \infty$, $n/(n-K)$ converges to 1 and the negative term in brackets converges to zero. What remains is

$$\frac{\epsilon'\epsilon}{n} = \frac{1}{n}\sum_{i=1}^{n} \epsilon_i^2 = \bar{\epsilon}^2$$

By the Markov Theorem,[7] $\mathrm{E}[\bar{\epsilon}^2] = \sigma^2$. Therefore, s^2 is consistent.

2.6 Statistical Inference

Assuming normally distributed errors, we have coefficients $\hat{\beta}_k$ on each of the k regressors has a normal distribution:

$$\hat{\beta}_k \sim N[\beta_k, \sigma^2 (X'X)^{-1}_{kk}]$$

Now, let S_{kk} be the kth diagonal of $(X'X)^{-1}$. Then, conditioned on X,

$$z_k = \frac{\hat{\beta}_k - \beta_k}{\sqrt{\sigma^2 S_{kk}}}$$

has a standard normal distribution. By using s^2 in the place of σ^2, we can derive a statistic analogous to z_k to test hypothesis about $\hat{\beta}_k$. For this, we use

$$\frac{(n-K)s^2}{\sigma^2} = \frac{e'e}{\sigma^2} = \left(\frac{\epsilon}{\sigma}\right)' M \left(\frac{\epsilon}{\sigma}\right)$$

which has a chi-squared distribution with $n-K$ degrees of freedom and is idependent of z_k.[8] Therefore, the ratio

$$t_k = \frac{(\hat{\beta}_k - \beta_k)/\sqrt{\sigma^2 S_{kk}}}{\sqrt{[(n-K)s^2/\sigma^2]/(n-K)}} = \frac{\hat{\beta}_k - \beta_k}{\sqrt{s^2 S_{kk}}} \tag{2.8}$$

has a t distribution with $n-K$ degrees of freedom and can be used to test hypothesis and form confidence intervals about elements of $\hat{\beta}$. A common test is whether a parameter β_k is statistically different from zero. The test statistic

$$t = \frac{\hat{\beta}_k}{s_{\hat{\beta}_k}}$$

[5] Those conditions are also known as "well-behaved data" and are usually met.
[6] Check Greene, page 66 for a detailed proof.
[7] Check Greene, page 1044.
[8] See Greene, 4.7.1 for details.

is the *t ratio* for $\hat{\beta}_k$. If $|\hat{\beta}_k|/s_{\hat{\beta}_k} > t_{\alpha/2}$, where $t_{\alpha/2}$ is the $100(1-\alpha)$ percent critical value from the t distribution with $n-K$ degrees of freedom, then the coefficient is said to be statistically significant. Accordingly, a confidence interval for β_k could be

$$Prob(\hat{\beta}_k - t_{\alpha/2}s_{\hat{\beta}_k} \leq \beta_k \leq \hat{\beta}_k + t_{\alpha/2}s_{\hat{\beta}_k}) = 1-\alpha \tag{2.9}$$

where $1-\alpha$ is the desired level of confidence.

2.7 Conditional Independence Assumption (CIA)

When regression coefficients can be interpreted as causal effects? Suppose we are studying the causal effect of going to college, indicated by the dummy c_i, on wages Y_i. Using the potential outcomes notation,[9] individual i has two potential outcomes: Y_{1i} if i goes to college ($c_i = 1$), or Y_{0i} otherwise ($c_i = 0$).

Comparisons of those who do and don't go to college is a poor measure of the causal effect of college:

$$\overbrace{\mathrm{E}[Y_i|c_i=1] - \mathrm{E}[Y_i|c_i=0]}^{\text{Observed difference in outcome}} = \underbrace{\mathrm{E}[Y_{1i} - Y_{0i}|c_i=1]}_{\text{Average treatment effect on the treated}} + \overbrace{\mathrm{E}[Y_{0i}|c_i=1] - \mathrm{E}[Y_{0i}|c_i=0]}^{\text{Selection bias}}$$

Therefore, if people who go to college have better potential outcomes than those who don't, selection bias is positive and the naive comparison $\mathrm{E}[Y_i|c_i=1] - \mathrm{E}[Y_i|c_i=0]$ exaggerates the effect of college. The *conditional independence assumption* (CIA) states that, conditional on a set of covariates X_i, potential outcomes are orthogonal to college attendance c_i – thus, selection bias disappears. Formally, the CIA requires

$$Y_{1i}\, Y_{0i} \perp c_i|X_i$$

With the CIA, condition-on-X_i comparisons of wages across school levels provides causal estimates of going to college:

$$\mathrm{E}[Y_i|X_i\, c_i=1] - \mathrm{E}[Y_i|X_i\, c_i=0] = \mathrm{E}[Y_{1i} - Y_{0i}|X_i]$$

With this framework, we go back to regression. Consider a bivariate regression model estimating the effect of treatment D_i on an outcome Y_i, with error term η_i:

$$Y_i = \alpha + \rho D_i + \eta_i$$

Suppose now the CIA holds given a vector of covariates X_i. We decompose the random part η_i into a function of observable characteristics and an error term: $\eta_i = X_i\gamma + \upsilon_i$. Since γ is defined by the regression of η_i on X_i, the residual υ_i and X_i are orthogonal by construction. Moreover, due to CIA the residual in

$$Y_i = \alpha + \rho D_i + X_i\gamma + \upsilon_i \tag{2.10}$$

is uncorrelated with the regressors D_i and X_i, and therefore the coefficient ρ is the causal effect of interest.

2.7.1 Omitted Variable Bias (OVB)

The CIA result makes it clear the importance of covariates in estimating causal effects. What if we do not include X_i when running 2.10? The *omitted variable bias* (OVB) formula states that

$$\frac{Cov\,(Y_i, D_i)}{Var\,(D_i)} = \rho + \gamma'\delta_{X,D} \tag{2.11}$$

where $\delta_{X,D}$ is the vector of coefficients from regressions of the elements of X_i on D_i. More generally, in matrix form, suppose that the true model is $Y = X_1\beta_1 + X_2\beta_2 + \epsilon$. The bias of the coefficient in the mispecified model $Y = X_1\tilde{\beta}_1 + \epsilon$ is given by

$$\begin{aligned}
\tilde{\beta}_1 &= (X_1'X_1)^{-1}X_1'Y = (X_1'X_1)^{-1}X_1'(X_1\beta_1 + X_2\beta_2 + \epsilon) \\
&= (X_1'X_1)^{-1}X_1'X_1\beta_1 + (X_1'X_1)^{-1}X_1'X_2\beta_2 + (X_1'X_1)^{-1}X_1'\epsilon \\
&= \beta_1 + (X_1'X_1)^{-1}X_1'X_2\beta_2 + (X_1'X_1)^{-1}X_1'\epsilon \\
\implies \mathrm{E}[\tilde{\beta}_1 - \beta_1|X] &= (X_1'X_1)^{-1}X_1'X_2\beta_2 \\
&= P_{1,2}\beta_2
\end{aligned}$$

[9] See Cunningham, chapter 3.

Hence, if one claims absence of OVB, this means that coefficients may be interpreted as causal. However, more covariates is not always better. For instance, *bad controls* – i.e., variables that are outcomes themselves, are affected by the variable interest, or are measured after the treatment – should not be included.

3 GLS and Heteroscedasticity

We now relax the homoscedasticity assumption **A4.** and allow the error's variance to vary across observations:

$$\mathrm{E}[\epsilon\epsilon'|X] = \sigma^2\Omega = \Sigma$$

where Ω is a positive definite matrix. Since **A3.** remains, the OLS estimator remains unbiased (see 2.5). However, the variance-covariance matrix of $\hat{\beta}_{OLS}$ under heteroscedastic errors is (using the expression in 2.6):

$$Var(\hat{\beta}_{OLS}|X) = (X'X)^{-1}X'\mathrm{E}[\epsilon\epsilon'|X]X(X'X)^{-1}$$

$$Var(\hat{\beta}_{OLS}|X) = \sigma^2(X'X)^{-1}(X'\Omega X)(X'X)^{-1} \tag{3.1}$$

Thus, under heteroscedasticity, OLS remains unbiased, consistent, and asymptotic normally distributed. However, since its variance is no longer $\sigma^2(X'X)^{-1}$, it is not efficient anymore. Additionally, the usual inference methods are no longer appropriate. Our approach in handling heteroscedasticity will depend on the structure of Ω, and whether we know it in the first place.

3.1 Generalized Least Squares (GLS)

We first assume Ω is known. We premultiply the original model $Y = X\beta + \epsilon$ by $\Omega^{-1/2}$:

$$\Omega^{-1/2}Y = \Omega^{-1/2}X\beta + \Omega^{-1/2}\epsilon$$
$$\tilde{Y} = \tilde{X}\beta + \tilde{\epsilon}$$

Note why we premultiply the model by $\Omega^{-1/2}$. Now, we have $\mathrm{E}[\tilde{\epsilon}\tilde{\epsilon}'|X] = \sigma^2$ – i.e., homoscedastic errors. Thus, the GLS estimator of β is

$$\hat{\beta}_{GLS} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}\tilde{Y}$$
$$= [(\Omega^{-1/2}X)'(\Omega^{-1/2}X)]^{-1}(\Omega^{-1/2}X)(\Omega^{-1/2}Y)$$

$$\hat{\beta}_{GLS} = (X'\Omega^{-1/2}X)^{-1}X'\Omega^{-1/2}Y \tag{3.2}$$

Under the model's assumptions, we have the properties:

1. if $\mathrm{E}[\tilde{\epsilon}|\tilde{X}] = 0$, $\hat{\beta}_{GLS}$ is unbiased;
2. if $p\lim_{n\to\infty}(\frac{1}{n})\tilde{X}'\tilde{X} = \tilde{Q}$ is a positive definite matrix, $\hat{\beta}_{GLS}$ is consistent;
3. $\hat{\beta}_{GLS} \sim \mathrm{N}[\beta, \sigma^2(X'\Omega^{-1/2}X)^{-1}]$;
4. $\hat{\beta}_{GLS}$ is BLUE in the generalized model.

3.1.1 Feasible Generalized Least Squares (FGLS)

Estimation of $\hat{\beta}_{GLS}$ is only possible with we know Ω. Suppose we know enough about Ω to assume it depends on a set of parameters θ: $\Omega = f(\theta)$. Then, if $\hat{\theta}$ is a *consistent* estimator of θ, the *feasible generalized least squares* estimator

$$\hat{\beta}_{FGLS} = (X'\hat{\Omega}^{-1/2}X)^{-1}X'\hat{\Omega}^{-1/2}Y \tag{3.3}$$

where $\hat{\Omega} = f(\hat{\theta})$ is asymptotically equivalent to the GLS estimator, i.e., it is efficient.[10]

[10] See Greene, page 156 for details.

3.2 Robust variance estimation

How bad is OLS when **A4.** is violated? The asymptotic variance-covariance matrix of $\hat{\beta}_{OLS}$ under heteroscedasticity is given by

$$\begin{aligned} Var(\hat{\beta}_{OLS}|X) &= \sigma^2(X'X)^{-1}(X'\Omega X)(X'X)^{-1} \\ &= (X'X)^{-1}\left(\sigma^2\sum_{i=1}^{n} w_i x_i x_i'\right)(X'X)^{-1} \end{aligned}$$

It follows that $\hat{\beta}_{OLS}$ is inefficient (larger variance) when compared to $\hat{\beta}_{GLS}$. In most cases, in which error's variances w_i depends on the variables of the model, the greater the dispersion of w_i across observations, larger the gap between $\hat{\beta}_{OLS}$ and $\hat{\beta}_{GLS}$. However, if the heteroscedasticity is uncorrelated with the model's variables X, then OLS is asymptotically reliable.[11]

3.2.1 White's consistent estimator

If Ω is known, GLS is an easy choice. If we have enough information about Ω to estimate it, we may use FGLS. However, using the wrong Ω or making untrue assumptions about its structure may be worse than the original heteroscedasticity problem. When uncertain, a third approach is to estimate an appropriate version of the OLS' variance-covariance matrix.

Remember from 3.1 that we seek an estimator for $Q = \frac{1}{n}\sum_i \sigma_i^2 x_i x_i'$. Since $\mathrm{E}[\epsilon^2] = \sigma^2$ and e is an estimator of ϵ, using OLS residuals is a natural choice. White (1980) has shown that under general assumptions, the estimator

$$S_0 = \frac{1}{n}\sum_i e_i^2 x_i x_i'$$

has $p\lim_{n\to\infty} S_0 = p\lim_{n\to\infty} \tilde{\mathbf{Q}}^*$. Thus, the *White heteroscedasticity consistent estimator* of $Var(\hat{\beta}_{OLS})$ is

$$V_\beta^{HC0} = (X'X)^{-1}\left(\sum_i e_i^2 x_i x_i'\right)(X'X)^{-1} \tag{3.4}$$

The White estimator 3.4 may be too optimistic in small samples because e_i^2 is biased towards zero. Making the same adjustment done for s^2 yields

$$V_\beta^{HC1} = \frac{n}{(n-K)}(X'X)^{-1}\left(\sum_i e_i^2 x_i x_i'\right)(X'X)^{-1}$$

Alternatively, one can use the standardized residuals $\bar{e}_i$ or the prediction errors $\tilde{e}_i$, what gives

$$\begin{aligned} V_\beta^{HC2} &= (X'X)^{-1}\left(\sum_i \bar{e}_i{}^2 x_i x_i'\right)(X'X)^{-1} \\ &= (X'X)^{-1}\left(\sum_i (1-h_{ii})^{-1} e_i^2 x_i x_i'\right)(X'X)^{-1} \end{aligned}$$

and

$$\begin{aligned} V_\beta^{HC3} &= (X'X)^{-1}\left(\sum_i \tilde{e}_i{}^2 x_i x_i'\right)(X'X)^{-1} \\ &= (X'X)^{-1}\left(\sum_i (1-h_{ii})^{-2} e_i^2 x_i x_i'\right)(X'X)^{-1} \end{aligned}$$

Finally,[12] note that

$$V_\beta^{HC3} \leq V_\beta^{HC2} \leq V_\beta^{HC0}$$

[11] See Greene, pages 159-162 for a detailed analysis of OLS' efficiency under heteroscedasticity.
[12] See Hansen, page 111 for a detailed discussion.

3.3 Tests for heteroscedasticity

It is desirable test general hypothesis concerning the existence and structure of the heteroscedasticity in a model. One simple framework is

$$H_0 : \sigma_i = \sigma^2, \quad \forall i;$$
$$H_1 : \text{Not } H_0$$

3.3.1 White's Test

White's test comes from the observation that, if there is no heteroscedasticity, the conventional estimator $V = s^2(X'X)^{-1}$ will be consistent for 3.1. It is carried out by obtaining R^2 for the regression of squared residuals e_i^2 on a constant and all covariates in X and all squares and cross products of covariates in X. The statistic is distributed as chi-squared with $P-1$ degrees of freedom, where P is the number of regressors in the equation, including the constant.

3.3.2 Breusch-Pagan's test

Breusch and Pagan have devised a Lagrange multiplier test of the hypothesis that $\sigma_i^2 = \sigma^2 f(\alpha_0 + \alpha z_i)$, where z_i is a vector of independent variables. The model is homoscedastic if $\alpha = 0$. The test can be carried out with a simple regression:

$$LM = \frac{1}{2}[g'Z(Z'Z)^{-1}Z'g]$$

where Z is the $n \times P$ matrix of observations on $(1, z_i)$, and g is the vector of observations of $g_i = \frac{e_i^2}{(e'e/n)-1}$. Under the null hypothesis of homoscedasticity, LM has a limiting chi-squared distribution with degrees of freedom equal to the number of variables in z_i.

4 Maximum Likelihood Estimator (MLE)

The pdf for a random variable y, conditioned on a set of parameters θ, is denoted by $f(y, \theta)$. The joint density of n i.i.d. observations from the data-generating process described by $f(y, \theta)$ is the product of the individual densities:

$$f(y_1, ..., y_n|\theta) = \prod_{i=1}^{n} f(y_i, \theta) = L(\theta, y) \tag{4.1}$$

The joint density 4.1 is the *likelihood function*. It is usually simpler to work with its log:

$$\ln L(\theta, y) = \sum_{i=1}^{n} \ln f(y_i, \theta) \tag{4.2}$$

The parameter vector of interest θ is identified (estimable) if for any other parameter vector, $\theta^* \neq \theta$, for some data y, $L(\theta^*|y) \neq L(\theta|y)$. The MLE $\hat{\theta}$ is the value of θ that maximizes 4.2, that is, it is the value for the parameter of interest that makes the observed sample y more probable. Under regularity,[13] the MLE is asymptotically consistent, normal, efficient, and invariant.

4.1 Classical Linear Regression Model

Take the classical linear model $y_i = x_i'\beta + \epsilon_i$. The (log) likelihood function for a sample of n i.i.d. disturbances is

$$L = (2\pi\sigma^2)^{n/2} \exp^{-\epsilon'\epsilon/(2\sigma^2)}$$
$$L = (2\pi\sigma^2)^{n/2} \exp^{-(y-X\beta)'(y-X\beta)/(2\sigma^2)}$$
$$\ln L = -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln\sigma^2 - \frac{(y-X\beta)'(y-X\beta)}{2\sigma^2}$$

The FOCs for maximizing the log-likelihood with respect to β are

$$\frac{\partial \ln L}{\partial \beta} = \frac{X'(y-X\beta)}{\sigma^2} = 0 \tag{4.3}$$
$$\frac{\partial \ln L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{(y-X\beta)'(y-X\beta)}{2\sigma^4} = 0 \tag{4.4}$$

Hence, we have

$$\hat{\beta}_{MLE} = (X'X)X'Y \tag{4.5}$$
$$\hat{\sigma}^2_{MLE} = \frac{e'e}{n} \tag{4.6}$$

The slope estimator is equal to the OLS estimator, whereas the variance differs by not making the correction for degrees of freedom. The Cramér-Rao bound for the variance of an unbiased estimator is the negative inverse of the expectation of

$$\begin{bmatrix} \frac{\partial^2 \ln L}{\partial\beta\partial\beta'} & \frac{\partial^2 \ln L}{\partial\beta\partial\sigma^2} \\ \frac{\partial^2 \ln L}{\partial\sigma^2\partial\beta'} & \frac{\partial^2 \ln L}{\partial^2(\sigma^2)^2} \end{bmatrix} = \begin{bmatrix} -\frac{X'X}{\sigma^2} & -\frac{X'\epsilon}{\sigma^4} \\ -\frac{\epsilon'X}{\sigma^4} & \frac{n}{2\sigma^4} - \frac{\epsilon'\epsilon}{\sigma^6} \end{bmatrix}$$

In taking expectations, the off-diagonal vanishes, leaving

$$\left[\mathbf{I}(\beta, \sigma^2)\right]^{-1} = \begin{bmatrix} \sigma^2(X'X)^{-1} & 0 \\ 0 & 2\sigma^4/n \end{bmatrix}$$

Therefore, the OLS slope estimator is the MLE for this model and inherits all the desirable asymptotic properties of ML estimators. Since 2.7 is unbiased, $\hat{\sigma}^2_{MLE}$ is biased towards zero in small samples. But since s^2 and $\hat{\sigma}^2_{MLE}$ differ only by a factor $-K/n$ this bias vanishes in large samples, thus both are asymptotically equivalent.

[13] See Greene, page 488.

4.2 The Generalized Regression Model

We now consider the generalized model from section 3: $y_i = x_i'\beta + \epsilon_i$ with $\mathrm{E}[\epsilon\epsilon'|X] = \sigma^2\Omega$. We first assume that Ω is a matrix of known constants. If the disturbances are multivariate normally distributed, the log-likelihood function for the sample is

$$\ln L = -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln\sigma^2 - \frac{(y - X\beta)'\Omega^{-1}(y - X\beta)}{2\sigma^2} - \frac{\ln|\Omega|}{2}$$

Since Ω is a matrix of known constants, the MLE of β is the vector that minimizes the generalized sum of squares: $S_*(\beta) = (y - X\beta)'\Omega^{-1}(y - X\beta)$. The FOCs are

$$\frac{\partial \ln L}{\partial \beta} = \frac{1}{\sigma^2}X'\Omega^{-1}(y - X\beta) = \frac{1}{\sigma^2}X_*'(y_* - X_*\beta) = 0$$

$$\frac{\partial \ln L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}(y - X\beta)'\Omega^{-1}(y - X\beta) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}(y_* - X_*\beta)'(y_* - X_*\beta) = 0$$

The solutions are the OLS estimators using the transformed data:

$$\hat{\beta}_{MLE} = (X_*'X_*)X_*'Y_* = (X'\Omega^{-1}X)X'\Omega^{-1}Y \tag{4.7}$$

$$\hat{\sigma}^2_{MLE} = \frac{1}{n}(Y_* - X_*\hat{\beta}_{MLE})'(Y_* - X_*\hat{\beta}_{MLE}) = \frac{1}{n}(Y - X\hat{\beta}_{MLE})'\Omega^{-1}(Y - X\hat{\beta}_{MLE}) \tag{4.8}$$

Therefore, with normally distributed disturbances, GLS is also ML.

5 IV and Endogeneity

Consider the case in which assumption **A3.** (no correlation between X and ϵ) is violated:

$$\mathrm{E}[\epsilon_i|x_i] = \eta_i$$

This implies that the independent variables in X are no longer exogenous:

$$\mathrm{E}[x_i\epsilon_i] = \gamma \neq 0$$

Remember from 2.5 that this means that OLS is not unbiased anymore. For example, take the model in 2.10, in which D_i is exogenous only conditioned on X_i:

$$Y_i = \alpha + \rho D_i + X_i\gamma + \upsilon_i$$

If X_i is not observable then D_i is endogenous and ρ would suffer from OVB. In this case, one can still estimate the causal estimand ρ using an instrumental variable z_i. The key is to z_i be correlated with D_i and orthogonal to any other determinants of Y_i, namely X_i and υ_i.[14] In this case, we have

$$\rho = \frac{Cov\ (Y_i, z_i)}{Cov\ (D_i, z_i)} = \frac{Cov\ (Y_i, z_i)/Var\ (z_i)}{Cov\ (D_i, z_i)/Var\ (z_i)}$$

The coefficient of interest ρ is the ratio of the coefficient of the regression of Y_i on z_i (the *reduced form*) to the coefficient of the regression of D_i on z_i (the *first stage*):

$$\text{First stage: } D_i = A_i\pi_{10} + \pi_{11}z_i + \xi_{1i} \tag{5.1}$$

$$\text{Reduced form: } Y_i = A_i\pi_{20} + \pi_{21}z_i + \xi_{2i} \tag{5.2}$$

where A_i is a set of control variables. The covariate-adjusted IV estimator is the sample analog of the ratio π_{21}/π_{11}:

$$\rho = \frac{\pi_{21}}{\pi_{11}} = \frac{Cov\ (Y_i, \tilde{z}_i)}{Cov\ (D_i, \tilde{z}_i)} \tag{5.3}$$

where $\tilde{z}_i$ is the residual from a regression of z_i on the exogenous covariates A_i. The formula in 5.3 is called the Indirect Least Squares (ILS) estimator of ρ in the structural equation

$$Y_i = \alpha A_i + \rho D_i + \eta_i$$

where $\eta_i = X_i\gamma + \upsilon_i$. Substituting the first stage 5.1, we obtain the reduced form 5.2:

$$\begin{aligned} Y_i &= \alpha A_i + \rho[A_i\pi_{10} + \pi_{11}z_i + \xi_{1i}] + \eta_i \\ &= A_i[\alpha + \rho\pi_{10}] + \rho\pi_{11}z_i + [\rho\xi_{1i} + \eta_i] \\ &= A_i\pi_{20} + \pi_{21}z_i + \xi_{2i} \end{aligned}$$

where $\pi_{20} \equiv \alpha + \rho\pi_{10}$, $\pi_{21} \equiv \rho\pi_{11}$, and $\xi_{21} \equiv \rho\xi_{1i} + \eta_i$. Also, we have

$$Y_i = \alpha A_i + \rho[A_i\pi_{10} + \pi_{11}z_i] + \xi_{2i}$$

where $[A_i\pi_{10} + \pi_{11}z_i]$ is the population fitted value of the first stage regression 5.1 of D_i on A_i and z_i. The fact that the coefficient on this fitted value equals ρ is because z_i and A_i are uncorrelated with the reduced form error, ξ_{2i}. Given a random sample, the first stage fitted values are estimated by

$$\hat{D}_i = A_i\hat{\pi}_{10} + \hat{\pi}_{11}z_i$$

where $\hat{\pi}_{10}$ and $\hat{\pi}_{11}$ are OLS estimates from the first stage regression 5.1. The coefficient on $\hat{D}_i$ on a regression of Y_i on A_i and $\hat{D}_i$ is the two-stage least squares (2SLS) estimator of ρ, i.e., is estimated with the *second stage* regression:

$$Y_i = \alpha A_i + \rho\hat{D}_i + [\eta_i + \rho(D_i - \hat{D}_i)] \tag{5.4}$$

Intuitively, conditional on covariates, 2SLS retains only the variation in D_i that is generated exogenously by the instrument z_i.

[14] This is known as the *exclusion restriction*, because z_i can be excluded from the model.

5.1 Example: measurement error

A common scenario in which endogeneity arises is when some variable is poorly measured. Take the true model $Y = \beta X + \epsilon$. The classical *errors-in-variables* model assumes that we observe

$$\tilde{x} = x + u$$
$$\mathrm{E}[u] = 0 \quad \mathrm{E}[Y'u] = 0 \quad \mathrm{E}[X'u] = 0 \quad \mathrm{E}[\epsilon' u] = 0$$

We have

$$y = \beta(\tilde{X} - u) + \epsilon = \beta\tilde{x} + (\epsilon - \beta u)$$

Since $\tilde{x}$ is correlated with u, there is now endogeneity bias. The OLS estimator is

$$\hat{\beta}_{OLS} = \frac{cov(\tilde{x}, y)}{var(\tilde{x})} = \frac{cov(x+u, \beta x + \epsilon)}{var(x+u)} \quad (*)$$
$$\implies p\lim_{n\to\infty} \hat{\beta}_{OLS} = \frac{\beta\sigma_x^2}{\sigma_x^2 + \sigma_u^2}$$

Therefore, the *attenuation bias* is

$$\hat{\beta}_{OLS} - \beta = -\frac{\sigma_u^2}{\sigma_x^2 + \sigma_u^2}\beta \tag{5.5}$$

Now, if u and x are correlated, (*) becomes

$$\hat{\beta}_{OLS} = \frac{cov(x+u, \beta x + \epsilon)}{var(x+u)} \implies p\lim_{n\to\infty} \hat{\beta}_{OLS} = \frac{\beta(\sigma_x^2 + \sigma_{xu})}{\sigma_x^2 + \sigma_u^2 + 2\sigma_{xu}}$$

Finally, the general case in which Y is also measured with error with observed $\tilde{y} = y + v$, where v can be correlated with x and/or y:

$$\hat{\beta}_{OLS} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{Y} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'(\tilde{X}\beta - U\beta + v + \epsilon)$$
$$= \beta + (\tilde{X}'\tilde{X})^{-1}\tilde{X}'(-U\beta + v + \epsilon)$$

Collecting the errors in a single matrix $W = [U|v]$:

$$p\lim_{n\to\infty} \hat{\beta}_{OLS} = \beta + p\lim_{n\to\infty} (\tilde{X}'\tilde{X})^{-1}\tilde{X}'W \begin{bmatrix} \beta \\ -1 \end{bmatrix}$$

So biases can always be seen in terms of the coefficients of regressing the measurement errors on the mismeasured variables.

5.2 Instrumental Variables (IV)

Suppose we have $k = 1, ..., K$ endogenous regressors X. The IV model assumes the existence of Z, a set o $l = 1, ..., L$ *instruments*, such that:

1. Z is *exogenous* (uncorrelated with ϵ): $p\lim_{n\to\infty}(\frac{1}{n})Z'\epsilon = 0$;
2. Z is *relevant* (correlated with X): $p\lim_{n\to\infty}(\frac{1}{n})ZX = Q_{ZX}$, where Q_{ZX} is an $L \times K$ full rank positive matrix;
3. Well-behaved data: $p\lim_{n\to\infty}(\frac{1}{n})Z'Z = Q_{ZZ}$, positive definite.

Since $\mathrm{E}[z_i\epsilon_i] = 0$,

$$p\lim\left(\frac{Z'\epsilon}{n}\right) = p\lim\left(\frac{Z'Y}{n}\right) - p\lim\left(\frac{Z'X\beta}{n}\right) = 0$$
$$\implies p\lim\left(\frac{Z'Y}{n}\right) = \left[p\lim\left(\frac{Z'X}{n}\right)\right]\beta$$

With L instruments for K endogenous regressors, if $L = K$ then $Z'X$ is a square matrix, and thus invertible:

$$\beta = \left[p\lim\left(\frac{Z'X}{n}\right)\right]^{-1} p\lim\left(\frac{Z'Y}{n}\right)$$

Thus, the IV estimator in the *just-identified* model ($l = k$) is

$$\hat{\beta}_{IV} = (Z'X)^{-1}Z'Y \tag{5.6}$$

with asymptotic covariance matrix given by

$$Var(\hat{\beta}_{IV}) = \sigma^2(Z'X)^{-1}(Z'Z)(X'Z)^{-1} \tag{5.7}$$

5.2.1 Wald estimator

Consider the simple case in which Z is a single dummy variable. We have

$$\begin{aligned} \mathrm{E}[Y|Z=1] &= \mathrm{E}[X|Z=1]\beta \\ \mathrm{E}[Y|Z=0] &= \mathrm{E}[X|Z=0]\beta \\ \implies \beta &= \frac{\mathrm{E}[Y|Z=1] - \mathrm{E}[Y|Z=0]}{\mathrm{E}[X|Z=1] - \mathrm{E}[X|Z=0]} \end{aligned}$$

Denoting the sample averages by

$$\bar{Y_1} = \frac{\sum_i Z_i Y_i}{\sum_i Z_i} \quad \bar{Y_0} = \frac{\sum_i (1-Z_i)Y_i}{\sum_i (1-Z_i)} \quad \bar{X_1} = \frac{\sum_i Z_i X_i}{\sum_i Z_i} \quad \bar{X_0} = \frac{\sum_i (1-Z_i)X_i}{\sum_i (1-Z_i)}$$

we get the Wald estimator for $\hat{\beta}_{IV}$:

$$\hat{\beta}_{Wald} = \frac{\bar{Y_1} - \bar{Y_0}}{\bar{X_1} - \bar{X_0}} \tag{5.8}$$

5.3 Two-Stages Least Squares (2SLS)

When $K < L$, the matrix $Z'X$ is not invertible. In this case, we use the projection of X into the column space of Z:

$$\hat{X} = Z(Z'Z)^{-1}Z'X$$
$$\hat{\beta}_{2SLS} = (\hat{X}'\hat{X})^{-1}\hat{X}'Y$$
$$\hat{\beta}_{2SLS} = [X'Z(Z'Z)^{-1}Z'X]^{-1}X'Z(Z'Z)^{-1}Z'Y \tag{5.9}$$

By multiplying out the matrices in the expression, you can show that

$$\hat{\beta}_{2SLS} = (X'(I - M_Z)X)^{-1}X'(I - M_Z)Y = (\hat{X}'\hat{X})^{-1}\hat{X}'Y$$

which implies the two-stage approach: *i.*) first stage: $X = Z\alpha + u$, and *ii.*) second stage: $Y = \hat{X}\beta + \epsilon$.

When $L = K$, we can show that $\hat{\beta}_{2SLS} = \hat{\beta}_{IV}$. Note that, if $K = L$, both $X'Z$ and $Z'X$ can be factored:

$$\begin{aligned} [X'Z(Z'Z)^{-1}Z'X] &= (Z'X)^{-1}[(Z'Z)^{-1}]^{-1}(X'Z) \\ &= (Z'X)^{-1}(Z'Z)(X'Z)^{-1} \end{aligned}$$

Thus,

$$\begin{aligned} \hat{\beta}_{2SLS} &= [X'Z(Z'Z)^{-1}Z'X]^{-1}X'Z(Z'Z)^{-1}Z'Y \\ &= (Z'X)^{-1}(Z'Z)(X'Z)^{-1}X'Z(Z'Z)^{-1}Z'Y \\ &= (Z'X)^{-1}Z'Y = \hat{\beta}_{IV} \end{aligned}$$

In the just identified model, both are also equal to the ILS estimator 5.3:

$$\begin{aligned} \hat{\beta}_{ILS} = \rho = \pi_{11}^{-1}\pi_{21} &= ((Z'Z)^{-1}(Z'X))^{-1}((Z'Z)^{-1}(Z'Y)) \\ &= (Z'X)^{-1}(Z'Z)(Z'Z)^{-1}(Z'Y) = (Z'X)^{-1}Z'Y = \hat{\beta}_{IV} \end{aligned}$$

5.3.1 Properties of 2SLS

Under the assumption that $\mathrm{E}[ZX']$ has full rank k and $\mathrm{E}[Z\epsilon] = 0$, the 2SLS estimator is consistent:

$$\begin{aligned}\hat{\beta}_{2SLS} &= (X'Z(Z'Z)^{-1}Z'X)^{-1}X'Z(Z'Z)^{-1}Z'(X\beta + \epsilon) \\ &= \beta + (X'Z(Z'Z)^{-1}Z'X)^{-1}X'Z(Z'Z)^{-1}Z'\epsilon\end{aligned}$$

Re-writing and applying the WLLN and CMT:

$$\begin{aligned}\hat{\beta}_{2SLS} - \hat{\beta}_{OLS} &= \left(\left(\frac{1}{n}X'Z\right)\left(\frac{1}{n}Z'Z\right)^{-1}\left(\frac{1}{n}Z'X\right)\right)^{-1} \times \left(\frac{1}{n}X'Z\right)\left(\frac{1}{n}Z'Z\right)^{-1}\left(\frac{1}{n}Z'\epsilon\right) \\ &\xrightarrow{p} (Q_{XZ}Q_{ZZ}^{-1}Q_{ZX})^{-1}Q_{XZ}Q_{ZZ}^{-1}\mathrm{E}[Z\epsilon] = 0\end{aligned}$$

Now, from the proof of consistency we derive the asymptotic distribution of $\hat{\beta}_{2SLS}$:

$$\sqrt{n}(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS}) = \left(\left(\frac{1}{n}X'Z\right)\left(\frac{1}{n}Z'Z\right)^{-1}\left(\frac{1}{n}Z'X\right)\right)^{-1} \times \left(\frac{1}{n}X'Z\right)\left(\frac{1}{n}Z'Z\right)^{-1}\left(\frac{1}{\sqrt{n}}Z'\epsilon\right)$$

By the CLT:

$$\frac{1}{\sqrt{n}}Z'\epsilon \xrightarrow{d} \mathrm{N}(0, \sigma^2\mathrm{E}[Z'Z])$$

Hence, we obtain

$$\sqrt{n}(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS}) \xrightarrow{d} \mathrm{N}(0, V_{\beta_{2SLS}})$$

where

$$V_{\beta_{2SLS}} = (Q_{XZ}Q_{ZZ}^{-1}Q_{ZX})^{-1}(Q_{XZ}Q_{ZZ}^{-1}\Omega Q_{ZZ}^{-1}Q_{XZ})(Q_{XZ}Q_{ZZ}^{-1}Q_{ZX})^{-1}$$

5.3.2 Bias in IV estimators

Suppose that the causal model of interest is $Y = \beta X + \epsilon$. Since X is endogenous, we use a $n \times q$ matrix of instrumental variables Z with the associated first-stage equation: $X = Z\pi + \mu$. The 2SLS estimator is

$$\hat{\beta}_{2SLS} = \beta + (X'P_ZX)^{-1}X'P_Z\epsilon$$

where $P_Z = Z(Z'Z)^{-1}X'Z'$ is the projection matrix that produces fitted values from a regression of X on Z. Solving, we have

$$\mathrm{E}[\hat{\beta}_{2SLS} - \beta] \approx [\mathrm{E}(\pi'Z'Z\pi) + \mathrm{E}(\mu'P_Z\mu)]^{-1}\mathrm{E}(\mu'P_Z\epsilon)$$

Therefore, the bias in 2SLS in not zero unless ϵ and μ are uncorrelated (what led us to use IV in the first place). Further manipulation of the above expression gives

$$\mathrm{E}[\hat{\beta}_{2SLS} - \beta] \approx \frac{\sigma_{\epsilon\mu}}{\sigma_\epsilon^2}\left[\frac{\mathrm{E}(\pi'Z'Z\pi)Q^{-1}}{\sigma_\epsilon^2} + 1\right]^{-1}$$

The first term in brackets is the F-statistic for the joint significance of all regressors in the first stage. Hence, we can write

$$\mathrm{E}[\hat{\beta}_{2SLS} - \beta] \approx \frac{\sigma_{\epsilon\mu}}{\sigma_\epsilon^2}\frac{1}{F+1}$$

Therefore, we can conclude:

1. as the first stage F-statistic gets small, the bias of 2SLS approaches $\frac{\sigma_{\epsilon\mu}}{\sigma_\epsilon^2}$;
2. 2SLS is centered on the same point as OLS when the first stage is zero;
   - 2SLS estimates are "biased towards" OLS estimates when the first stage is weak.
3. The bias of 2SLS vanishes when F gets large, as it should happen in large samples when $\pi \neq 0$[15]

[15] In small samples, one alternative is the Jackknife IV estimator (JIVE). See Hansen, 12.20.

5.4 Hausman and Wu specification test

Under the hypothesis that both the OLS and IV estimators are consistent, we can test whether X is endogenous by comparing both estimators:

$$d = \beta_{IV} - \beta_{OLS}$$

We might test the hypothesis that $d = 0$ with a Wald statistic:

$$H = \frac{d'[(\hat{X}'\hat{X})^{-1} - (X'X)^{-1}]^{-1}d}{s^2}$$

That is equivalent to:

$$Y = X\beta + \hat{X}\gamma + \epsilon$$

If X is not endogenous, then γ should be zero.

5.5 Weak Instruments

The use of instrumental variables is threatened if the instruments are *weak* – that is, if the correlation between Z and X is small. Consider the simplest case where $k = l = 1$:

$$\begin{aligned} Y &= X\beta + \epsilon \\ X &= Z\gamma + u \end{aligned}$$

Thus, β is identified if and only if $\gamma \neq 0$, which occurs when $\mathrm{E}[XZ] \neq 0$. Suppose this condition fails, therefore $\mathrm{E}[XZ] = 0$ and $\gamma = 0$.[16] What happens to the distribution of OLS and IV estimators of β? For simplicity we assume conditional homoscedasticity and normalize variances to unity. Thus

$$var\left[\left(\begin{array}{c} \epsilon \\ u \end{array}\middle| Z\right)\right] = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

The errors have non-zero correlation $\rho \neq 0$ when the variables are endogenous. By the CLT we have the joint convergence

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\begin{pmatrix} Z_i\epsilon_i \\ Z_i u_i \end{pmatrix} \xrightarrow{d} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} \sim \mathrm{N}\left(0, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right)$$

It is useful to define $\xi_0 = \xi_1 - \rho\xi_2$ which is normal and independent of ξ_2. As a benchmark, note that the OLS estimator satisfies

$$\hat{\beta}_{OLS} - \beta = \frac{n^{-1}\sum_{i=1}^{n} u_i\epsilon_i}{n^{-1}\sum_{i=1}^{n} u_i^2} \xrightarrow{p} \rho \neq 0$$

so endogeneity causes $\hat{\beta}_{OLS}$ to be inconsistent. Under identification failure $\gamma = 0$, the asymptotic distribution of the IV estimator is

$$\hat{\beta}_{IV} - \beta = \frac{\frac{1}{\sqrt{n}}\sum_{i=1}^{n} Z_i\epsilon_i}{\frac{1}{\sqrt{n}}\sum_{i=1}^{n} Z_i X_i} \xrightarrow{d} \frac{\xi_1}{\xi_2} = \rho + \frac{\xi_0}{\xi_2}$$

Hence, $\hat{\beta}_{IV}$ does not converge in probability to a limit, rather it converges in distribution to a random variable. Thus, the estimator is inconsistent.

5.6 Local Average Treatment Effects (LATE)

Consider a case that both the endogenous regressor X and the instrument Z are binary variables. The dependent variable is a function of X and an unobservable vector u, $Y = h(X, u)$, and the endogenous regressor X is a function of instrument Z and u : $X = g(Z, u)$.

[16] This is the case of full identification failure. Weak instrument relates to situations in which γ is small, but not zero. Check Hansen, 12.35-36 for details.

Given a realization Z the observables are $X = X(Z)$ and $Y = Y(X)$. In this model, the effect of X on Y is $C = Y(1) - Y(0)$. A common feature of interest is the average treatment effect (ATE):

$$ATE = \mathrm{E}[C] = \mathrm{E}[Y(1) - Y(0)] \tag{5.10}$$

However, to estimate the ATE with endogenous X require strong assumptions, such that the causal effect C is constant across individuals. One particular feature that can be estimated is the local average treatment effect (LATE), or the average effect upon those affected by the instrument. Given the unobservable u, the regressor X is a function of a realization of Z. Take the Card (1995) example. Proximity to College ($Z = 1$) can have an impact on individual decision on education (X), but not necessarily is a one-to-one relation. In short, there are four types of people:

	$X(0) = 0$	$X(0) = 1$
$X(1) = 0$	Never takers	Deniers
$X(1) = 1$	Compliers	Always takers

To identify the parameters of interest we need to assume that there are no Deniers, e.g. people who go to college only if they do not live close to one. That is the monotonicity condition ($X(1) \geq X(0)$). We also need other three standard IV assumptions: independence between Y and Z, the exclusion restriction (Z affects Y only through D), and relevance (first-stage). The average causal effect (ACE) is the expectation 5.10 – to estimate it, we need to observe some individuals that go to college and some who don't. That is unfeasible for the Never takers and Always takers groups. The LATE is defined as the ACE for the Compliers:

$$LATE = \frac{\mathrm{E}[Y|Z = 1] - \mathrm{E}[Y|Z = 0]}{\mathrm{E}[X|Z = 1] - \mathrm{E}[X|Z = 0]} \tag{5.11}$$

Note that LATE equals the Wald estimator 5.8 for the slope coefficient in the IV model. However, the LATE framework makes it clear that IV estimates should be interpreted as causal effects only for the population of compliers. When causal effects are heterogeneous, a different choice of instrument will yield a different LATE since the population of compliers would also likely change.

5.7 Generalized Method of Moments (GMM)

Let $g_i(\beta)$ be a known $l \times 1$ function of the ith observation and a $k \times 1$ parameter β. A moment equation model is summarized by the moment equations

$$\mathrm{E}[g_i(\beta)] = 0$$

For instance, in the IV model $g_i(\beta) = Z_i(Y_i - X_i'\beta)$. The methods of moments estimator (MME) is the parameter value which sets the sample analog equal to zero:

$$\bar{g}_n(\beta) = \frac{1}{n}\sum_{i=1}^{n} Z_i(Y_i - X_i'\beta) = \frac{1}{n}(Z'Y - Z'X\beta) \tag{5.12}$$

However, when the model is overidentified ($l > k$) this is not possible since there are more equations than parameters. One solution is to find an estimator which makes $\bar{g}_n(\beta)$ as close to zero as possible. Define the vector $\mu = Z'Y$, the matrix $G = Z'X$ and the error $\eta = \mu - G\beta$. Thus, we can write 5.12 as $\mu = G\beta + \eta$. Hence, we can simply regress μ on G, obtaining $\hat{\beta}$ – which minimizes the sum of squared errors, $\eta'\eta$. Generally, the GMM estimator is obtained via weighted least squares (allowing non-homogeneous errors). For some weighting matrix W:

$$\hat{\beta}_{GMM} = (G'WG)^{-1}(G'W\mu) = (X'ZWZ'X)^{-1}(X'ZWZ'Y) \tag{5.13}$$

Note that when the model is just-identified ($l = k$) then $X'Z$ is $k \times k$ so

$$\hat{\beta}_{GMM} = (Z'X)^{-1}W^{-1}(X'Z)^{-1}(X'ZWZ'Y) = (Z'X)^{-1}Z'Y = \hat{\beta}_{IV}$$

Also, when $W = (Z'Z)^{-1}$, the GMM and 2SLS estimators are equal.

5.7.1 Distribution of GMM estimator

Let $Q = \mathrm{E}[ZX']$ and $\Omega = \mathrm{E}[ZZ'\epsilon^2]$. Then

$$\left(\frac{1}{n}X'Z\right)W\left(\frac{1}{n}Z'X\right) \xrightarrow{p} Q'WQ$$

and

$$\left(\frac{1}{n}X'Z\right)W\left(\frac{1}{n}Z'\epsilon\right) \xrightarrow{d} Q'W\mathrm{N}(0,\Omega)$$

Therefore, as $n \to \infty$, $\sqrt{n}(\hat{\beta}_{GMM} - \beta) \xrightarrow{d} \mathrm{N}(0, V_{\beta_{GMM}})$, where

$$V_{\beta_{GMM}} = (Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1}$$

5.7.2 Efficient GMM

The asymptotically optimal weight matrix W_0 is one which minimizes $V_{\beta_{GMM}}$. It can be shown that efficient estimator uses $W_0 = \Omega^{-1}$:

$$\hat{\beta}_{GMM} = (X'Z\Omega Z'X)^{-1}(X'Z\Omega Z'Y) \tag{5.14}$$

Therefore, the asymptotic distribution of the efficient estimator is

$$\sqrt{n}(\hat{\beta}_{GMM} - \beta) \xrightarrow{d} \mathrm{N}(0, (Q'\Omega^{-1}Q)^{-1})$$

Also, for any $W > 0$,

$$\begin{gathered}(Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1} - (Q'\Omega^{-1}Q)^{-1} > 0 \\ \implies Var(\hat{\beta}_{GMM}) \leq Var(\tilde{\beta}_{GMM})\end{gathered}$$

To construct the efficient GMM estimator we need a consistent estimator $\hat{W}$ of $W_0 = \Omega^{-1}$. The usual is to form an estimator $\hat{\Omega}$ of Ω and set $\hat{W} = \hat{\Omega}^{-1}$. The *two-step GMM estimator* uses $\hat{\beta}_{2SLS}$ as a one-step consistent estimator of β to construct the weight matrix estimator $\hat{W}$. Set $\tilde{\epsilon}_i = Y_i - X_i'\hat{\beta}_{2SLS}$ and $\tilde{g}_i = g(\tilde{\beta}) = Z_i\tilde{\epsilon}_i$. The moment estimator of Ω is then

$$\hat{\Omega} = \frac{1}{n}\sum_{i=1}^{n}\tilde{g}_i\tilde{g}_i' = \frac{1}{n}Z'\epsilon\epsilon'Z \tag{5.15}$$

Recall that the 2SLS estimator is the GMM estimator given the weight matrix $\hat{W} = (Z'Z)^{-1}$. This is asymptotically equivalent to use $W = (\mathrm{E}[ZZ'])^{-1}$. In contrast, the efficient weight matrix takes the form $W = (\mathrm{E}[ZZ'\epsilon^2])^{-1}$. If errors are homoscedastic, the efficient matrix is $(\mathrm{E}[ZZ'])^{-1}\sigma^{-2}$, which is equivalent to the 2SLS matrix since scaling doesn't matter. Therefore, under homoscedasticity 2SLS and efficient GMM are equally efficient. However, under heteroscedastic errors 2SLS is inefficient relative to the optimal GMM.

6 Panel Data

Panel data follows individuals $i = 1, ..., N$ during time periods $t = 1, ..., T$. The general framework for panel data is a regression model of the form

$$\begin{aligned} y_{it} &= x_{it}'\beta + z_i'\alpha + \epsilon_{it} \\ &= x_{it}'\beta + c_i + \epsilon_{it} \end{aligned}$$

There are K regressors in x_{it}, not including a constant term. The heterogeneity, or individual effect, is $z_i'\alpha$ where z_i contains a constant term and a set of fixed individual characteristics that can be observed (race, gender, location, etc) or unobserved (skill, preferences).

6.1 Pooled Regression

The simplest model in panel regression is pooled regression, in which z_i has only a constant term:

$$y_{it} = x_{it}'\beta + \epsilon_{it} \tag{6.1}$$

where we assume $\mathrm{E}[X_{it}\epsilon_{it}] = 0$. The standard estimator of β in the pooled regression model is OLS:

$$\begin{aligned} \hat{\beta}_{pool} &= \left(\sum_{i=1}^{n}\sum_{t\in T} X_{it}'X_{it}\right)^{-1}\left(\sum_{i=1}^{n}\sum_{t\in T} X_{it}'Y_{it}\right) \\ &= \left(\sum_{i=1}^{n} X_i'X_i\right)^{-1}\left(\sum_{i=1}^{n} X_i'Y_i\right) \\ &= (X'X)^{-1}X'Y \end{aligned}$$

The pooled regression model is ideally suited for a context where the errors ϵ_{it} satisfy strict mean independence: $\mathrm{E}[\epsilon_{it}|X_i] = 0$. Under this assumption $\hat{\beta}_{pool}$ is unbiased for β:

$$\begin{aligned} \hat{\beta}_{pool} &= \left(\sum_{i=1}^{n} X_i'X_i\right)^{-1}\left(\sum_{i=1}^{n} X_i'(X_i\beta + \epsilon_i)\right) \\ &= \beta + \left(\sum_{i=1}^{n} X_i'X_i\right)^{-1}\left(\sum_{i=1}^{n} X_i'\epsilon_i\right) \\ \mathrm{E}[\hat{\beta}_{pool}|X] &= \beta \end{aligned}$$

Under the additional assumption that ϵ_{it} are uncorrelated and homoscedastic the covariance estimator takes the classical form. In general, however, we expect errors to be correlated across time t for a given individual. The conventional solution is to use the heteroscedasticity-robust covariance matrix estimator in the same fashion as White's:

$$V_{pool} = (X'X)^{-1}\left(\sum_{i=1}^{n} X_i'e_ie_i'X_i\right)(X'X)^{-1}$$

6.1.1 The within- and between- groups estimators

We can formulate the pooled regression model in three ways. First, the original formulation 6.1. The other two are in terms of group means,

$$\bar{y}_i = \bar{x}_i'\beta + \bar{\epsilon}_i \tag{6.2}$$

or deviations from the group means

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)'\beta + \epsilon_i - \bar{\epsilon}_i \tag{6.3}$$

All three could be consistently estimated with OLS. Consider the matrices of sums of squares and cross products that would be used in each case. In 6.1, we would have variation about the overall means:

$$S_{XX}^{total} = \sum_{i=1}^{n}\sum_{t\in T}(x_{it} - \bar{\bar{x}})(x_{it} - \bar{\bar{x}})' \text{ and } S_{XY}^{total} = \sum_{i=1}^{n}\sum_{t\in T}(x_{it} - \bar{\bar{x}})(y_{it} - \bar{\bar{y}})'$$

In 6.3 the moment matrices are *within-groups*, i.e., variation around group means:

$$S_{XX}^{within} = \sum_{i=1}^{n}\sum_{t\in T}(x_{it}-\bar{x}_i)(x_{it}-\bar{x}_i)' \text{ and } S_{XY}^{within} = \sum_{i=1}^{n}\sum_{t\in T}(x_{it}-\bar{x}_i)(y_{it}-\bar{y}_i)'$$

Finally, in 6.2 the moments matrices are *between-groups*, i.e., variation of the group means around the overall means:

$$S_{XX}^{between} = \sum_{i=1}^{n}\sum_{t\in T}(\bar{x}_i-\bar{\bar{x}})(\bar{x}_i-\bar{\bar{x}})' \text{ and } S_{XY}^{between} = \sum_{i=1}^{n}\sum_{t\in T}(\bar{x}_i-\bar{\bar{x}})(\bar{y}_i-\bar{\bar{y}})'$$

It is easy to show that $S_{XX}^{total} = S_{XX}^{within} + S_{XX}^{between}$ and $S_{XY}^{total} = S_{XY}^{within} + S_{XY}^{between}$. Therefore, the three possible OLS estimators of β are

$$\beta^{total} = [S_{XX}^{total}]^{-1}S_{XY}^{total} = [S_{XX}^{within} + S_{XX}^{between}]^{-1}[S_{XY}^{within} + S_{XY}^{between}] \tag{6.4}$$

$$\beta^{within} = [S_{XX}^{within}]^{-1}S_{XY}^{within} \tag{6.5}$$

$$\beta^{between} = [S_{XX}^{between}]^{-1}S_{XY}^{between} \tag{6.6}$$

Finally, rearranging the terms we can write the least squares estimator as a matrix weighted average of the within- and between- groups estimators:

$$\beta^{total} = F^{within}\beta^{within} + F^{between}\beta^{between}$$

where

$$F^{within} = [S_{XX}^{within} + S_{XX}^{between}]^{-1}[S_{XY}^{within} = I - F^{between}$$

6.2 Fixed Effects

The fixed effects model arises from the assumption that the time-invariant, individual-specific unobserved effect c_i is correlated with regressors x_{it}:

$$y_{it} = x_{it}'\beta + c_i + \epsilon_{it}$$

When c_i and X_{it} are correlated the pooled estimator is biased. The fixed effects approach sets $\mathrm{E}[c_i|X_i] = h(X_i)$, which yields

$$\begin{aligned} y_{it} &= x_{it}'\beta + h(X_i) + \epsilon_{it} + [c_i - h(X_i)] \\ &= x_{it}'\beta + \alpha_i + \epsilon_{it} + [c_i - h(X_i)] \\ &= x_{it}'\beta + \alpha_i + \epsilon_{it} \end{aligned}$$

With the assumption that $Var[c_i|X_i]$ is constant, we have a classical linear regression model. The fixed effects formulation implies that differences across groups can be captured in differences in the constant term.

6.2.1 Least squares estimation

Let y_i be the T observations for the ith unit, $\mathbf{1}_i$ be a $T \times 1$ column of ones, and let ϵ_i be the associated vector of disturbances. Then,

$$y_i = X_i\beta + \mathbf{1}_i\alpha_i + \epsilon_i$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}\beta + \begin{bmatrix} \mathbf{1}_1 & 0 & \dots & 0 \\ 0 & \mathbf{1}_2 & \dots & 0 \\ & & \vdots & \\ 0 & 0 & \dots & \mathbf{1}_n \end{bmatrix}\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$$y = [X \quad \mathbf{d}_1 \quad \mathbf{d}_2 \quad \dots \quad \mathbf{d}_n]\begin{bmatrix} \beta \\ \alpha \end{bmatrix} + \epsilon$$

where $\mathbf{d}_i$ is a dummy variable indicating the ith unit. Let the $nT \times n$ matrix $D = [\mathbf{d}_1 \quad \mathbf{d}_2 \quad ... \quad \mathbf{d}_n]$. Then, assembling all nT rows gives

$$y = X\beta + D\alpha + \epsilon \tag{6.7}$$

The model in 6.7 is known as *least squares dummy variable* (LSDV) model. By the FWL theorem, the LSDV estimator for β and estimated residuals $\hat{\epsilon}$ can be obtained from the regression of the residuals from the regression of Y on D on the residuals of the regression of X on D. Or,

$$\hat{\beta}_{fe} = (X'M_DX)^{-1}(X'M_DY) \tag{6.8}$$

where

$$M_D = I - D(D'D)^{-1}D'$$

Note that $M_D = \text{diag}\{M^i, ..., M^n\}$, with

$$M^i = I_i - \frac{1}{T}\mathbf{1}_i\mathbf{1}_i'$$

is the individual-specific demeaning operator: any $T \times 1$ vector z_i multiplied by M^i creates $M^i z_i = z_i - \bar{z}\mathbf{1}_i$. Therefore, $M_DX = \bar{X}$ and $M_DY = \bar{Y}$, and the fixed effects estimator is equivalent to a regression of $(y_{it} - \bar{y}_i)$ on $(x_{it} - \bar{x}_i)$. Denoting $\dot{X}_i = M_iX_i = X_{it} - \bar{X}_i$ and $\dot{Y}_i$ equivalently we have:

$$\begin{aligned}
\hat{\beta}_{fe} &= \left(\sum_{i=1}^{n}\sum_{t\in T}\dot{X}_{it}\dot{X}_{it}'\right)^{-1}\left(\sum_{i=1}^{n}\sum_{t\in T}\dot{X}_{it}\dot{Y}_{it}\right) \\
&= \left(\sum_{i=1}^{n}\dot{X}_{it}{}'\dot{X}_{it}\right)^{-1}\left(\sum_{i=1}^{n}\dot{X}_{it}{}'\dot{Y}_{it}\right) \\
&= \left(\sum_{i=1}^{n}X_i'M_iX_i\right)^{-1}\left(\sum_{i=1}^{n}X_i'M_iY_i\right) \\
&= (X'M_DX)^{-1}(X'M_DY)
\end{aligned}$$

Under strict mean independence $\text{E}[\epsilon_{it}|X_i] = 0$, $\hat{\beta}_{fe}$ is unbiased:

$$\begin{aligned}
\hat{\beta}_{fe} - \beta &= \left(\sum_{i=1}^{n}X_i'M_iX_i\right)^{-1}\left(\sum_{i=1}^{n}X_i'M_i\epsilon_i\right) \\
\implies \text{E}[\hat{\beta}_{fe} - \beta|X] &= \left(\sum_{i=1}^{n}X_i'M_iX_i\right)^{-1}\left(\sum_{i=1}^{n}X_i'M_i\text{E}[\epsilon_i|X]\right) = 0
\end{aligned}$$

Let $\Sigma_i = \text{E}[\epsilon_i\epsilon_i'|X_i]$. The covariance matrix of the fixed effects estimator is

$$V_{fe} = \left(\sum_{i=1}^{n}\dot{X}_i{}'\dot{X}_i\right)^{-1}\left(\sum_{i=1}^{n}\dot{X}_i{}'\Sigma_i\dot{X}_i\right)\left(\sum_{i=1}^{n}\dot{X}_i{}'\dot{X}_i\right)^{-1}$$

When errors are homoscedastic and serially uncorrelated, this reduces to

$$V_{fe} = \sigma^2\left(\sum_{i=1}^{n}\dot{X}_i{}'\dot{X}_i\right)^{-1} = \sigma^2(X'M_DX)^{-1} = \sigma^2[S_{XX}^{within}]^{-1}$$

where the disturbance variance estimator is

$$\hat{\sigma}^2 = \frac{1}{nT-n-K}\sum_{i=1}^{n}\sum_{t\in T}e_{it}^2 = \frac{(M_DY - M_DX\hat{\beta}_{fe})'(M_DY - M_DX\hat{\beta}_{fe})}{nT-n-K} \tag{6.9}$$

Finally, note that

$$V_{fe} = \sigma^2\left(\sum_{i=1}^{n}\dot{X}_i{}'\dot{X}_i\right)^{-1} \geq \sigma^2\left(\sum_{i=1}^{n}X_i'X_i\right)^{-1} = V_{pool}$$

6.2.2 Time effects

The identifying assumption of the fixed effects model is that regressors are strictly exogenous conditional on the unobserved effect ($\mathrm{E}[\epsilon_{it}|X, c_i] = 0$. A typical violation is common shocks that affect all units in the same way and are correlated with x_{it}.

We can allow for such common shocks by including time effects into the model. For example, we can add a linear time trend common to all units:

$$y_{it} = x'_{it}\beta + \alpha_i + t + \epsilon_{it}$$

The LSDV approach can be extended to include a time fixed effects as well, that is, common shocks in each time period. The model is obtained by including additional $T-1$ dummy variables (one must be dropped to avoid perfect collinearity):

$$y_{it} = x'_{it}\beta + \alpha_i + \delta_t + \epsilon_{it}$$

However, this formulation is asymmetric since each of the group effects is a group-specific intercept, whereas the time effects are contrasts (comparisons to the excluded base period). A symmetric form is

$$y_{it} = x'_{it}\beta + \mu + \alpha_i + \delta_t + \epsilon_{it}$$

where a full n and T effects are included but we impose $\sum_i \alpha_i = \sum_i \delta_t = 0$. Least squares estimates are obtained by regressing

$$y^*_{it} = y_{it} - \bar{y}_i - \bar{y}_t + \bar{\bar{y}}$$

on

$$x^*_{it} = x_{it} - \bar{x}_i - \bar{x}_t + \bar{\bar{x}}$$

where the group-specific, period-specific, and overall means are

$$\bar{y}_i = \frac{1}{T}\sum_{t\in T} y_{it},\ \bar{y}_t = \frac{1}{n}\sum_{i=1}^{n} y_{it},\ \text{and}\ \bar{\bar{y}} = \frac{1}{nT}\sum_{i=1}^{n}\sum_{t\in T} y_{it}$$

and likewise for x_{it}. Finally, we can also add time fixed effects and unit-specific linear time trends:

$$y_{it} = x'_{it}\beta + \alpha_i + \delta_t + t \times g_i + \epsilon_{it}$$

6.3 Random effects

Consider the reformulation of the model

$$y_{it} = x'_{it}\beta + (\alpha + u_i) + \epsilon_{it}$$

where there are K regressors including a constant and now the single constant term is the mean of the unobserved heterogeneity, $\mathrm{E}[z'_i\alpha]$. The component u_i is a random heterogeneity specific to unit i constant through time. The random effects model is characterized by its set of assumptions, specifically, that u_i and ϵ_{it} are conditionally mean zero, uncorrelated, and homoscedastic:

$$\begin{gathered}\mathrm{E}[\epsilon_{it}|X] = 0, \quad \mathrm{E}[\epsilon^2_{it}|X] = 0, \quad \mathrm{E}[\epsilon_{it}\epsilon_{jt}|X] = 0, \\ \mathrm{E}[u_i|X] = 0, \quad \mathrm{E}[u^2_i|X] = 0, \quad \mathrm{E}[u_i\epsilon_{it}|X] = 0\end{gathered}$$

This implies that the vector of errors e_i for unit i has the covariance structure

$$\begin{aligned}\mathrm{E}[e_i e'_i|X] &= \begin{bmatrix} \sigma^2_u + \sigma^2_\epsilon & \sigma^2_u & \dots & \sigma^2_u \\ \sigma^2_u & \sigma^2_u + \sigma^2_\epsilon & \dots & \sigma^2_u \\ \vdots & \vdots & \ddots & \vdots \\ \sigma^2_u & \sigma^2_u & \dots & \sigma^2_u + \sigma^2_\epsilon \end{bmatrix} \\ &= \sigma^2_\epsilon \Omega_i\end{aligned}$$

Note that $\Omega_i = I_i + \mathbf{1}_i\mathbf{1}_i'\sigma_u^2/\sigma_\epsilon^2$. The matrix Ω_i is not random, but it depends on i since its dimension depends on the number of observed time periods T_i. Given the error structure, the natural estimator for β is GLS:

$$\begin{aligned}\hat{\beta}_{re} &= (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}Y \\ &= \left(\sum_{i=1}^n X_i'\Omega_i X_i\right)^{-1}\left(\sum_{i=1}^n X_i'\Omega_i Y_i\right) \\ &= \left(\sum_{i=1}^n \tilde{X}_i{}'\tilde{X}_i\right)^{-1}\left(\sum_{i=1}^n \tilde{X}_i{}'\tilde{X}_i\right)\end{aligned}$$

Where $\tilde{X}_i = \Omega_i^{-1/2}X_i$ and $\tilde{Y}_i = \Omega_i^{-1/2}Y_i$. A feasible GLS estimator replaces Ω_i with an estimator. Under the model's assumptions, $\hat{\beta}_{re}$ is unbiased for β:

$$\begin{aligned}\hat{\beta}_{re} - \beta &= \left(\sum_{i=1}^n X_i'\Omega_i^{-1}X_i\right)^{-1}\left(\sum_{i=1}^n X_i'\Omega_i^{-1}\epsilon_i\right) \\ \implies \mathrm{E}[\hat{\beta}_{re} - \beta|X] &= \left(\sum_{i=1}^n X_i'\Omega_i^{-1}X_i\right)^{-1}\left(\sum_{i=1}^n X_i'\Omega_i^{-1}\mathrm{E}[\epsilon_i|X]\right) = 0\end{aligned}$$

The conditional variance of β_{re} is

$$V_{re} = \left(\sum_{i=1}^n X_i'\Omega_i^{-1}X_i\right)^{-1} \leq \left(\sum_{i=1}^n X_i'X_i\right)^{-1}\left(\sum_{i=1}^n X_i'\Omega_i^{-1}X_i\right)^{-1}\left(\sum_{i=1}^n X_i'X_i\right)^{-1} = V_{pool}$$

and thus the random effects estimator is more efficient than the pooled estimator. Note, however, that the two variance matrices are identical when there is no individual specific effect ($\sigma_u^2 = 0$): $V_{pool} = V_{re} = \left(\sum_{i=1}^n X_i'X_i\right)^{-1}\sigma_\epsilon^2$.

6.3.1 Feasible GLS

Define $P_i = \mathbf{1}_i(\mathbf{1}_i'\mathbf{1}_i)^{-1}\mathbf{1}_i$ so that $M_i = I_i - P_i$. Thus while M_i is the within operator, P_i is the individual mean operator since $P_iY_i = \mathbf{1}_i\bar{Y}_i$. We can write

$$\Omega_i = I_i + \mathbf{1}_i'\mathbf{1}_i\sigma_u^2/\sigma_\epsilon^2 = I_i + \frac{T_i\sigma_u^2}{\sigma_\epsilon^2}P_i = M_i + \theta_i^{-2}P_i$$

where

$$\theta_i = \frac{\sigma_\epsilon}{\sqrt{\sigma_\epsilon^2 + T_i\sigma_u^2}}$$

Since M_i and P_i are idempotent and orthogonal we find that

$$\Omega_i^{-1/2} = M_i + \theta_i P_i = I_i - (1-\theta_i)\mathbf{1}_i P_i$$

Therefore, the transformation used by the GLS estimator is

$$\tilde{Y}_i = (I_i - (1-\theta_i)P_i)Y_i = Y_i - (1-\theta_i)\mathbf{1}_i\bar{Y}_i$$

Hence, the GLS estimator for the random effects model depends on θ_i, which in turn depends on σ_ϵ^2 and σ_u^2. The feasible GLS estimator is obtained by estimating $\hat{\theta}_i$ using $\hat{\sigma}_\epsilon^2$ as in 6.9 and $\hat{\sigma}_u^2$ as

$$\hat{\sigma}_u^2 = \max\left[0, \hat{\sigma}_b^2 - \frac{\hat{\sigma}_\epsilon^2}{T}\right]$$

where $\hat{\sigma}_b^2 = \frac{1}{N-k}\sum_{i=1}^N e_{bi}^2$ is the estimator for the variance of ϵ_{bi}, the disturbances of the between estimator.[17]

Finally, the feasible random effects GLS estimator is $\beta_{FGLS}^{re} = (\tilde{X}_i'\tilde{X}_i)^{-1}\tilde{X}_i'\tilde{Y}_i$, where

$$\begin{aligned}\tilde{Y}_i &= Y_i - (1-\hat{\theta}_i)\mathbf{1}_i\bar{Y}_i \\ \tilde{X}_i &= X_i - (1-\hat{\theta}_i)\mathbf{1}_i\bar{X}_i\end{aligned}$$

[17] See Hansen, 17.15.

6.4 Differences-in-Differences

Diff-in-Diff is characterized by two periods (*pre-* and *post*-treatment) and two groups (*treated* and *control*). The post period is identified by the $Post_t$ dummy variable and the treatment group is identified by the $Treat_i$ dummy. The standard specification is given by

$$y_{it} = \beta_0 + \beta_1 Post_t + \beta_2 Treat_i + \delta Post_t \times Treat_i + \epsilon_{it} \tag{6.10}$$

The differences-in-differences estimator δ gives the average treatment effect. We have

	Before	After	After - Before
Control	β_0	$\beta_0 + \beta_1$	β_1
Treated	$\beta_0 + \beta_2$	$\beta_0 + \beta_1 + \beta_2 + \delta$	$\beta_1 + \delta$
Control - Treated	β_2	$\beta_2 + \delta$	δ

Thus, δ is obtained by getting rid of the initial difference between treated and control groups (β_2) and the difference between pre and post periods common to both groups (β_1).

The fundamental assumption for identification is that the treatment $D_{it} = Post_t \times Treat_i$ is independent of errors ϵ_{it}. This means that all other changes coincidental with D have no effect on the outcome Y. This assumption is known as "parallel trends assumption": absent of treatment D, control and treated units would have followed the same trends.

In practice, we may have more than two periods, the identifying assumption could be valid conditional on some covariates, and time and unit fixed effects are useful. Hence, we can generalize 6.10 to

$$y_{it} = \beta_0 + \delta Post_t \times Treat_i + \mu X_{it} + \alpha_i + \lambda_i + \epsilon_{it}$$

where α_i are unit fixed effects, λ_i are time fixed effects, and X_{it} is a set of controls.

6.4.1 Event Studies and pre-trends

It is impossible to test the parallel trends assumption, but we can check pre-trends. If trends are similar before policy, then it is likely that they would have continued to be similar in the post-period absent a real treatment. We do this by estimating δ for each period k before and after policy adoption in period t:

$$y_{it} = \beta_0 + \sum_{k \leq t} \delta_k^{pre} D_k^{pre} \times Treat_i + \sum_{k > t} \delta_k^{post} D_k^{post} \times Treat_i + \mu X_{it} + \alpha_i + \lambda_i + \epsilon_{it}$$

if parallel trends holds, then $\delta_k^{pre} = 0$ for all $k \leq t$.

6.5 Clustering

The correct standard errors of $\hat{\beta}$ are obtained with the default covariance matrix when errors ϵ are i.i.d., or with the heteroscedastic-robust methods when errors are not independent. Clustering deals with the common case in panel data in which errors are correlated *within* cluster but independent *across* clusters. Consider a simple bivariate model estimated in a data with group structure:

$$y_{ig} = \beta_0 + \beta_1 x_g + \epsilon_{ig}$$

where y_{ig} is the dependent variable of individual i in group (cluster) g, and x_g is the regressor of interest (which varies only at the group level). In this case it is prudent to assume equicorrelation within groups, that is, for individuals i and j in the same group g:

$$\mathrm{E}[\epsilon_{ig}\epsilon_{jg}] = \rho_\epsilon \sigma_\epsilon^2 > 0$$

where ρ_ϵ is the intergroup error correlation and σ_ϵ^2 is the error variance. This comes from a random effects model assuming that the error has a group structure:

$$\epsilon_{ig} = \upsilon_g + \eta_{ig}$$

where υ_g is a group-specific random component and η_{ig} is a mean-zero individual error term uncorrelated across individuals. Give the error structure, we have

$$\rho_\epsilon = \frac{\sigma_\upsilon^2}{\sigma_\upsilon^2 + \sigma_\eta^2}$$

where σ_υ^2 and σ_η^2 are the variances of υ and η, respectively. The term ρ_ϵ is also called the *interclass correlation coefficient*. Now, let $V_c(\hat{\beta}_1)$ be the conventional OLS variance for $\hat{\beta}_1$, while $V(\hat{\beta}_1)$ is the correct variance given the grouped error structure. With nonstochastic regressors fixed at the group level and groups of equal size n, we have[18]

$$\tau = \frac{V(\hat{\beta}_1)}{V_c(\hat{\beta}_1)} = 1 + (n-1)\rho_\epsilon$$

Hence, ignoring interclass correlation results in overestimating precision. Conventional standard errors become increasingly misleading as n and ρ_ϵ increase. Since variation occurs across groups, adding extra observation do not provide new independent information. With fixed sample size, larger groups (bigger n) mean fewer clusters, i.e., less variation. More generally, allowing the regressor x_{ig} to vary at the individual level and for different group sizes n_g, the Moulton factor τ is given by

$$\tau = \frac{V(\hat{\beta}_1)}{V_c(\hat{\beta}_1)} = \sqrt{1 + \left[\frac{V(n_g)}{\bar{n}} + \bar{n} - 1\right]\rho_x\rho_\epsilon} \tag{6.11}$$

where $\bar{n}$ is average group size and ρ_x is the interclass correlation if x_{ig}. Thus, clustering has a bigger impact when ρ_x is large – the worse case is when the regressor is fixed within groups ($\rho_x = 1$).

A covariance matrix estimation that allows for heteroscedasticity and autocorrelation is the cluster-robust covariance matrix estimator (CRVE), clustered by group:

$$\hat{V}_{CR} = (X'X)^{-1}\left(\sum_{g=1}^{G} X_g' e_g e_g' X_g\right)(X'X)^{-1} \tag{6.12}$$

where $e_g = Y_g - X_g\hat{\beta}_1$ are the within-group residuals. Alternatively, CRVE can be constructed using cluster-level prediction errors such as $\tilde{e}_g = Y_g - X_g\hat{\beta}_{(-g)}$, where $\hat{\beta}_{(-g)}$ is the OLS estimator omitting cluster g:

$$\hat{V}_{CR3} = (X'X)^{-1}\left(\sum_{g=1}^{G} X_g' \tilde{e}_g \tilde{e}_g' X_g\right)(X'X)^{-1}$$

Finally, another possibility is the cluster-robust Jackknife estimator:

$$\hat{V}_{CR}^{jack} = \frac{G-1}{G}\sum_{g=1}^{G}(\hat{\beta}_{(-g)} - \bar{\beta})(\hat{\beta}_{(-g)} - \bar{\beta})'$$

where $\bar{\beta} = G^{-1}\sum_{g=1}^{G}\hat{\beta}_{(-g)}$.

6.5.1 Feasible GLS

There a potential efficiency gains for FGLS compared to OLS. We first specify a model for the within-cluster equicorrelation, $\Omega_g = \mathrm{E}[\epsilon_g\epsilon_g'|X_g]$. Given a consistent estimator $\hat{\Omega}_g \xrightarrow{p} \Omega_g$, the FGLS estimator of β is

$$\hat{\beta}_{FGLS} = \left(\sum_{g=1}^{G} X_g'\hat{\Omega}_g^{-1}X_g\right)^{-1}\sum_{g=1}^{G} X_g'\hat{\Omega}_g^{-1}Y_g$$

and the cluster-robust variance is

$$\hat{V}_{CR}(\hat{\beta}_{FGLS}) = (X'\hat{\Omega}^{-1}X)^{-1}\left(\sum_{g=1}^{G} X_g'\hat{\Omega}_g^{-1}e_g e_g'\hat{\Omega}_g^{-1}X_g\right)(X'\hat{\Omega}^{-1}X)$$

where $e_g = Y_g - X_g\hat{\beta}_{FGLS}$ and ϵ_g and ϵ_j are assumed to be uncorrelated for $g \neq j$.

[18] See MHE chapter 8 appendix for a derivation.

6.5.2 Cluster Bootstrap

Bootstrapping is a resampling method that offers an alternative to inference based on asymptotic formulas. Basically, it consists in repeatedly drawing from the sample with replacement and, at each draw, obtaining an estimator for the parameter of interest. Suppose we will make B bootstrap samples. At each draw b, we would

1. form G clusters $(y_1, X_1), \ldots, (y_G, X_G)$ by resampling with replacement G times from the original sample of clusters;
2. estimate $\hat{\beta}_b$ in the bth bootstrap sample

Then, we can compute the variance of the B estimates as

$$\hat{V}_{CR}^{boot} = \frac{1}{B-1} \sum_{b=1}^{B} (\hat{\beta}_b - \bar{\hat{\beta}}_b)(\hat{\beta}_b - \bar{\hat{\beta}}_b)'$$

where $\bar{\hat{\beta}}_b$ is the mean $\hat{\beta}_b$ over the B bootstrap samples.

6.5.3 Multi-way clustering

Cluster-robust variance matrix estimates are of the form $V(\hat{\beta}) = (X'X)^{-1}\hat{B}(X'X)^{-1}$. For one-way clustering with groups $g = 1, \ldots, G$ we can write

$$\hat{B} = \sum_{i=1}^{N} \sum_{j=1}^{N} x_i x_j' e_i e_j \mathbf{1}[i, j \text{ in same cluster } g]$$

Now, for two-way clustering with groups $g = 1, \ldots, G$ and $h = 1, \ldots, H$:

$$\hat{B} = \sum_{i=1}^{N} \sum_{j=1}^{N} x_i x_j' e_i e_j \mathbf{1}[i, j \text{ share any of the two clusters}] =$$

$$\sum_{i=1}^{N} \sum_{j=1}^{N} x_i x_j' e_i e_j \mathbf{1}[i, j \text{ in same cluster } g] + \sum_{i=1}^{N} \sum_{j=1}^{N} x_i x_j' e_i e_j \mathbf{1}[i, j \text{ in same cluster } h]$$

$$- \sum_{i=1}^{N} \sum_{j=1}^{N} x_i x_j' e_i e_j \mathbf{1}[i, j \text{ in both clusters }]$$

Intuitively, two-way clustering is obtained by computing three different cluster-robust variance matrices (the first dimension, the second dimension, and the intersection of the two). The last one is subtracted to avoid double-counting.

6.5.4 Inference with few clusters

The CRVE assumes $G \to \infty$. However, from 6.11 we know that the overestimation of precision is larger when the number of clusters is small. When that is the case, that are some possible solutions. We focus on one-way clustering and the Wald t-statistic $w = (\hat{\beta} - \beta_0)/s_{\hat{\beta}}$.

- Bias-corrected CRVE:
  - Simplest option is to use CRVE with the rescaled error $\tilde{e}_g = \sqrt{c} e_g$, where $c = \frac{G}{G-1}$.
  - CR2VE generalizes HC2, with $\tilde{e}_g^{CR2} = [I_{N_g} - H_{gg}]^{-1/2} e_g$ where $H_{gg} = X_g(X'X^{-1}X_g$.
  - CR3VE generalizes HC3, with $\tilde{e}_g^{CR3} = \sqrt{G/(G-1)}[I_{N_g} - H_{gg}]^{-1/2} e_g$, which is the same as jackknife.
- Wild Cluster Bootstrap, following the steps:
  1. Obtain the OLS estimator $\hat{\beta}$ and residuals e_g, $g = 1, \ldots, G$.

2. For each cluster g form $e_g^* = e_g$ or $e_g^* = -e_g$ with probability 0.5, and form $\hat{y}_g^* = X_g'\hat{\beta} + e_g^*$. Iterate over this step and obtain B bootstrap resamples $\{(\hat{y}_1^*, X_1), \ldots, (\hat{y}_g^*, X_g)\}_b$.
3. Calculate the OLS estimate $\hat{\beta}_{1,b}^*$ and standard error $s_{\hat{\beta}_{1,b}^*}$.
4. Form the Wald test statistic $w_b^* = (\hat{\beta}_{1,b}^* - \hat{\beta}_{1,b})/s_{\hat{\beta}_{1,b}^*}$. Reject H_0 at level α if and only if $w < w_{[\alpha/2]}^*$ or $w > w_{[1-\alpha/2]}^*$, where $w_{[q]}^*$ denotes the qth quantile of $w_1^*, \ldots, w_B^*$.

7 Nonparametric Methods

We now turn to the estimation of the conditional expectation function of the type

$$\mathrm{E}[Y|X = x] = m(x)$$

where the function $m(x)$ can take any nonlinear shape. The nonparametric regression model with a real-valued regressor is

$$Y = m(x) + \epsilon \tag{7.1}$$

in which we assume $\mathrm{E}[\epsilon|X] = 0$, $\mathrm{E}[\epsilon\epsilon'|X] = \sigma^2(X)$, and continuity of $m(x)$. The goal is usually to estimate $m(x)$ at a single point or at a set of points.

7.1 Estimation

7.1.1 Binned Means Estimator

Since $m(x)$ is continuous, it should be possible to get a approximation of $m(x)$ by taking the average of the observations for which X_i is close to x, for instance for observations for which $|X_i - x| \leq h$ for some small $h > 0$. In this case, h is a *bandwidth*. This *binned means estimator* can be written as

$$\hat{m}(x) = \frac{\sum_{i=1}^n \mathbf{1}\{|X_i - x| \leq h\} Y_i}{\sum_{i=1}^n \mathbf{1}\{|X_i - x| \leq h\}} \tag{7.2}$$

The binned estimator $\hat{m}(x)$ is the step function that is constant within each bin $X = x$ and equals the binned mean.

7.1.2 Kernel Regression

One problem with 7.2 is that it is always a step function of x because the weights are discontinuous indicator functions. In order to obtain a continuous $\hat{m}(x)$ we need to use continuous weights. A generalization of 7.2 is obtained by replacing the indicator function with a kernel function:[19]

$$\hat{m}_{nw}(x) = \frac{\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) Y_i}{\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right)} \tag{7.3}$$

The estimator 7.3 is called *kernel regression* or *Nadaraya-Watson* estimator. Larger values of h will result in estimates smoother in x (as $h \to \infty$, $\hat{m}_{nw}(x)$ reaches the sample mean $\bar{Y}$), and smaller values of h will yield more erratic estimates (in the limit, with $h \to 0$, $\hat{m}_{nw}(x)$ is simply the scatter plot of Y_i on X_i).

7.1.3 Local Linear Estimator

The Nadaraya-Watson estimator (NW) solves the minimization problem

$$\hat{m}_{nw}(x) = \arg\min_m \sum_{i=1}^n K\left(\frac{X_i - x}{h}\right)(Y_i - m)^2$$

which means that it is making the local approximation $m(X) \simeq m(x)$ for $X \simeq x$ and is therefore also called *local constant* estimator. An alternative approximation is the *local linear* (LL). The LL estimator uses the approximation $m(X) \simeq m(x) + m'(x)(X - x) + \epsilon$. The approximating model can be written as

$$Y \simeq Z(X, x)'\beta(x) + \epsilon$$

where $\beta(x) = (m(x), m'(x))'$ and $Z(X, x) = \begin{pmatrix} 1 \\ X - x \end{pmatrix}$. This is a linear rergession with regressor vector $Z_i(x)$ and coefficient vector $\beta(x)$. The LL estimator is thus obtained via weighted least squares with the kernel weights:

$$\hat{\beta}_{LL} = \left(\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) Z_i(x) Z_i(x)'\right)^{-1} \left(\sum_{i=1}^n K\left(\frac{X_i - x}{h}\right) Z_i(x) Y_i\right) = (Z'KZ)^{-1}(Z'KY) \tag{7.4}$$

[19] Check Hansen, 19.3 for a definition of kernel functions.

7.1.4 Local Polynomial Estimator

The NW and LL estimators are both special cases of the *local polynomial* (LP) estimator. The idea is to approximate $m(x)$ by a polynomial of fixed degree p, and then estimate locally using kernel weights. The approximation model is a pth order Taylor series approximation

$$\begin{aligned} Y = m(X) + \epsilon \simeq m(x) + m'(X-x) + \cdots + m^{(p)}(x)\frac{(X-x)^p}{p!} + \epsilon \\ = Z(X,x)'\beta(x) + \epsilon_i \end{aligned}$$

where

$$Z(X,x) = \begin{pmatrix} 1 \\ X-x \\ \vdots \\ \frac{(X-x)^p}{p!} \end{pmatrix} \quad \beta(x) = \begin{pmatrix} m(x) \\ m'(x) \\ \vdots \\ m^{(p)}(x) \end{pmatrix}$$

Hence, the estimator is

$$\hat{\beta}_{LP} = \left(\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right) Z_i(x) Z_i(x)'\right)^{-1} \left(\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right) Z_i(x) Y_i\right) = (Z'KZ)^{-1}(Z'KY) \tag{7.5}$$

The NW estimator is obtained with $p = 0$, and the LL estimator with $p = 1$. There is a trade-off between polynomial order p and local smoothing bandwidth h: increasing p improves model approximation and thereby we can use a larger h, but increasing p also leads to larger estimation variance.

7.2 Asymptotic Bias

Since $\mathrm{E}[Y|X = x] = m(x)$, the conditional mean of the NW estimator is

$$\mathrm{E}[\hat{m}_{nw}(x)|X] = \frac{\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right)\mathrm{E}[Y_i|X_i]}{\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right)} = \frac{\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right) m(X_i)}{\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right)} \tag{7.6}$$

Assuming some regularity conditions,[20] we can show that

$$\mathrm{E}[\hat{m}_{nw}(x)|X] = m(x) + h^2 B_{nw}(x) + o_p(h^2) + O_p\left(\sqrt{\frac{h}{n}}\right), \text{ where } B_{nw} = \frac{1}{2}m''(x) + f(x)^{-1}f'(x)m'(x) \tag{7.7}$$

$$\mathrm{E}[\hat{m}_{LL}(x)|X] = m(x) + h^2 B_{LL}(x) + o_p(h^2) + O_p\left(\sqrt{\frac{h}{n}}\right), \text{ where } B_{LL} = \frac{1}{2}m''(x) \tag{7.8}$$

in which $f(x)$ denotes the marginal density of X. The terms $h^2 B_{nw}(x)$ and $h^2 B_{LL}(x)$ are called the *asymptotic bias* of these estimators, which is proportional to the degree of smoothing h^2. Local averaging smooths $m(x)$ and induces bias, and this *smoothing bias* is increasing in the curvature of $m(x)$. This bias is a by-product of nonparametric estimation and can only be reduced by using a small bandwidth – which, in turn, leads to higher estimation variance.

7.3 Asymptotic Variance

From 7.6, we deduce that

$$\hat{m}_{nw}(x) - \mathrm{E}[\hat{m}_{nw}(x)|X] = \frac{\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right)\epsilon_i}{\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right)}$$

Hence, we can calculate the finite sample variance of $\hat{m}_{nw}(x)$ as

$$var(\hat{m}_{nw}(x)|X) = \frac{\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right)^2 \sigma^2(X_i)}{\left(\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right)\right)^2} \tag{7.9}$$

[20] Check Hansen, 19.6.

Under the same regularity conditions as before, we can show that

$$var(\hat{m}_{nw}(x)|X) = var(\hat{m}_{LL}(x)|X) = \frac{R_K \sigma^2(x)}{f(x)nh} + o_p\left(\frac{1}{nh}\right) \tag{7.10}$$

where $R_K = \int_{-\infty}^{\infty} K(u)^2 du$ is the *roughness* of the kernel $K(u)$. This reflects the fact that those are local estimators. The precision of $\hat{m}(x)$ is low for regions where ϵ has a large conditional variance and/or X has a low density (few observations).

7.4 Bandwidth Selection

7.4.1 AIMSE

We can define the asymptotic MSE of $\hat{m}(x)$ as the sum of the squared asymptotic bias and asymptotic variance:

$$AMSE[x] \equiv h^4 B(x)^2 + \frac{R_K \sigma^2}{nhf(x)}$$

A global measure of fit can be obtained by integrating $AMSE[x]$ weighted by $f(x)w(x)$ for some integrable weight function $w(x)$, which is called the asymptotic integrated MSE ($AIMSE$):

$$AIMSE = h^4 \bar{B} + \frac{R_K}{nh}\bar{\sigma}^2$$

where $\bar{B}$ and $\bar{\sigma}^2$ are the integration (weighted by $f(x)w(x)$) of B and σ^2 over S, the support of X (the region where $f(x) > 0$). Thus, one term of $AIMSE$ is increasing in the bandwidth h and the other is decreasing. The bandwidth that minimizes the AIMSE is

$$h_0 = \left(\frac{R_K \bar{\sigma}^2}{4\bar{B}}\right)^{1/5} n^{-1/5} \simeq 0.58\left(\frac{\bar{\sigma}^2}{n\bar{B}}\right)^{1/5} \tag{7.11}$$

7.4.2 Rule of Thumb (ROT)

A reference approach can be used to derive a rule-of-thumb bandwidth for the local linear estimator. First, set $w(x) = \mathbf{1}[\xi_1 \leq x \leq \xi_2]$ and form a preliminary estimator of the regression function $m(x)$ using a q-order polynomial:

$$m(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_q x^q$$

The coefficient estimates $\hat{\beta}_0, \ldots, \hat{\beta}_q$ are obtained by least squares, which we use to compute the second order derivative $\hat{m}''(x) = 2\hat{\beta}_2 + 6\hat{\beta}_3 x + \cdots + q(q-1)\hat{\beta}_q x^{q-2}$. Then, we write $\bar{B}$ as the expectation

$$\bar{B} = \mathrm{E}[B(X)^2 w(X)] = \mathrm{E}\left[\left(\frac{1}{2}m''(X)\right)^2 \mathbf{1}[\xi_1 \leq X \leq \xi_2]\right]$$

and form the moment estimator

$$\hat{B} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{1}{2}m''(X)\right)^2 \mathbf{1}[\xi_1 \leq X \leq \xi_2]$$

We assume that errors are homoscedastic so that $\bar{\sigma}^2 = \sigma^2(\xi_2 - \xi_1)$ and estimate σ^2 by $\hat{\sigma}^2$ from the preliminary regression. Plugging this into 7.11 gives the reference bandwidth

$$h_{rot} = 0.58\left(\frac{\hat{\sigma}^2(\xi_2 - \xi_1)}{n\hat{B}}\right)^{1/5} \tag{7.12}$$

7.4.3 Cross-Validation

The idea behind selecting bandwidths via cross-validation is to estimate the model fit based on leave-one-out estimation. Ideally, we would like to select h to minimize the integrated mean-squared error (IMSE) of $\hat{m}(x,h)$ as estimate of $m(x)$:

$$IMSE_n(h) = \int_S \mathrm{E}[(\hat{m}(x,h) - m(x))^2] f(x) w(x) dx$$

The weight function $w(x)$ can be omitted when X has bounded support and this is typically done in practice. We estimate the difference $\hat{m}(x,h) - m(x)$ at $x = X_i$ by leave-one-out prediction errors:

$$\tilde{e}_i = Y_i - \tilde{m}_{-i}(X_i, h)$$

A reasonable estimator of $IMSE_n(h)$ is the weighted average mean squared prediction errors

$$CV(h) = \frac{1}{n} \sum_{i=1}^{n} \tilde{e}_i h^2 w(X_i) \tag{7.13}$$

The function $CV(h)$ is known as *cross-validation criterion* and is essentially an unbiased estimator of $IMSE_n(h) + \bar{\sigma}^2$. Hence, it is recommended to select h as the value which minimizes $CV(h)$:

$$h_{cv} = \arg\min_{h \geq h_l} CV(h) \tag{7.14}$$

where the restriction $h \geq h_l$ for some $h_l > 0$ imposes that the function is not evaluated over very small bandwidths.

7.5 Variance Estimation

The NW, LL, and LP estimators can be written as

$$\hat{\beta}(x) = (Z'KZ)^{-1}(Z'KY) = (Z'KZ)^{-1}(Z'Km) + (Z'KZ)^{-1}(Z'K\epsilon)$$

where m is the $n \times 1$ vector of means $m(X_i)$. Thus, conditionally on the regressors X:

$$V_{\hat{\beta}}(x) = var(\hat{\beta}|X) = (Z'KZ)^{-1}(Z'KDKZ)(Z'KZ)^{-1} \tag{7.15}$$

where $D = diag(\sigma^2(X_1), \ldots, \sigma^2(X_n))$. A White-type estimator can be formed by replacing $\sigma^2(X_i)$ with squared residuals e_i^2 or prediction errors $\tilde{e}_i^2$:

$$\hat{V}_{\hat{\beta}}(x) = (Z'KZ)^{-1}\left(\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right)^2 Z_i(x) Z_i(x)' \tilde{e}_i^2\right)(Z'KZ)^{-1}$$

7.6 Regression Discontinuity (RD)

A common application of nonparametric methods is *regression discontinuity designs* (RDD). A *sharp RDD* is characterized by a treatment that is determined by a threshold function of X, usually $T = \mathbf{1}[X \geq c]$. The covariate X is called the *running variable* and c is the *cut-off*.

7.6.1 Identification

Let θ denote the random individual treatment effect and $\theta(x) = \mathrm{E}[\theta|X = x]$ the conditional ATE. Set $\bar{\theta} = \theta(c)$, the conditional ATE for the subpopulation affected at the margin by the (assumed) exogenous decision to set the cut-off at c. Define $m(x) = \mathrm{E}[Y|X = x]$, $m(x+) = \lim_{z \downarrow x}(z)$, and $m(x-) = \lim_{z \uparrow x}(z)$. Under the assumptions that the conditional expectation of the untreated and treated outcomes are continuously affected by the running variable, $\bar{\theta}$ is identified by

$$\bar{\theta} = m(c+) - m(c-) \tag{7.16}$$

Hence, the RDD identifies the conditional ATE only at the treatment cut-off. Since non-linearity may be misled with a discontinuity around the cutoff, $\bar{\theta}$ must be estimated via nonparametric methods.

7.6.2 Estimation

The conditional ATE can be calculated from the conditional mean $m(x)$. Estimation of the conditional mean non-parametrically allowing for discontinuity is the same as separately estimating the conditional mean for untreated observations $x_i < c$ and treated ones $x_i \geq c$. The estimator for $\bar{\theta}$ is the difference between the two.

Local linear estimators are usually employed in RDD. The LL estimator for $x < c$ is

$$\hat{\beta}_0(x) = \left(\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right) Z_i(x) Z_i(x)' \mathbf{1}[X_i < c] \right)^{-1} \left(\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right) Z_i(x) Y_i \mathbf{1}[X_i < c] \right)$$

and for $x \geq c$ is

$$\hat{\beta}_1(x) = \left(\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right) Z_i(x) Z_i(x)' \mathbf{1}[X_i \geq c] \right)^{-1} \left(\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right) Z_i(x) Y_i \mathbf{1}[X_i \geq c] \right)$$

The estimator of the conditional mean is the first element of the coefficient vectors: $\hat{m}(x) = [\hat{\beta}_0(x)]_1 \mathbf{1}[x < c] + [\hat{\beta}_1(x)]_1 \mathbf{1}[x \geq c]$. The estimator of $\bar{\theta}$ is the difference at c:

$$\hat{\theta} = [\hat{\beta}_1(c)]_1 - [\hat{\beta}_0(c)]_1 = \hat{m}(c+) - \hat{m}(c-) \tag{7.17}$$

A simple RDD estimator can be obtained via the regression

$$Y = \beta_0 + \beta_1 X + \beta_2 (X - c) T + \theta T + \epsilon$$

for the subpopulation of observations such that $|X - c| \geq h$.

7.6.3 Fuzzy RDD

In a *fuzzy RDD* the cut-off does not perfectly separates treated from untreated, but the conditional probability of treatment, $p(x) = \mathrm{P}[T = 1 | X = x]$, is discontinuous at the cut-off. Defining the left and right limits at c as $p(c+)$ and $p(c-)$, the fuzzy RDD applies when $p(c+) \neq p(c-)$. Under the further assumption that T is independent of θ for X near c, it can be shown that

$$\bar{\theta} = \frac{m(c+) - m(c-)}{p(c+) - p(c-)}$$

As before, the estimator for discontinuity $m(c+) - m(c-)$ is obtained by local linear regression of Y on X on the two sides of the cut-off, and $p(c+) - p(c-)$ is similarly estimated by local linear regressions of Y on T, yielding

$$\hat{\theta} = \frac{\hat{m}(c+) - \hat{m}(c-)}{\hat{p}(c+) - \hat{p}(c-)}$$